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Substitution effect of Asian economies on China's industrial and supply chains: from the perspective of global production network

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This paper views China Production Network (CPN) as an integral component of the Global Production Network (GPN) and constructs null model (GIVCBN-I model) and the counterfactual model (GIVCBN-II or GIVCBN-III model) to identify and measure the potential industrial relocation risk exposure of CPN based on the Multi-Regional Input-Output (MRIO) table. The main findings are as follows. Firstly, Altasia will induce more of China's industrial and supply chains to break compared with ASEAN, which also means that strong industrial complementarity exists between Japan, South Korea, India, Bangladesh and the ASEAN member countries, enabling Altasia to have a more significant substitution effect on China. Secondly, the counterfactual models' network-level characteristic indicators are worse than those of the null model in economic terms, suggesting that removing trade barriers for intermediate goods within Altasia could lead to the decoupling of industrial sectors in the CPN, thereby accelerating the trend of China's industrial and supply chains relocating offshore. Thirdly, according to the comparison results of node-level characteristic indicators, Altasia has weakened China's influence scope, profitability, and robustness of risk within the global industrial and supply chains, but mainly concentrating on its resource-intensive and labor-intensive sectors rather than capital-intensive and technology-intensive sectors, which indicates that some China's industrial sectors still maintain substantial competitive advantages in the GPN. In sum, this paper provides theoretical guidance for identifying and analyzing the trends of industrial relocation in the Asia-Pacific region and helps industrial policymakers deepen the understanding of regional economic integration and its impact.

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Introduction

Profound adjustments have been observed in the global industrial division of labor in recent years, with new trends such as inward contraction, localized relocation, and regional clustering. The global stage is witnessing a fierce competition among economies for dominance in the industrial and supply chains. Some economies have even resorted to political and economic measures to promote re-shoring, near-shoring, and friend-shoring of their industrial layouts, underlining the urgency and importance of this competition. According to a report by The Economist in February 2023, fourteen Asian economies are on the verge of potentially supplanting China as the focal point of the global industrial and supply chains amidst escalating tensions between China and the United States. These economies, collectively referred to as *Alternative Asian Supply Chain*, or the so-called *Altasia*, include the vast majority of the *Association of Southeast Asian Nations (ASEAN)* except Myanmar, as well as Japan and South Korea in East Asia, and India and Bangladesh in South Asia.

Importance of Altasia. The Economist suggested that Altasia could collectively rival or surpass China regarding economic and manufacturing capabilities. However, this issue needs to be explored from multiple perspectives. On the one hand, China has become a global production center after more than three decades of effort. It is not realistic for Altasia to replace China overnight. On the other hand, as China's demographic dividend fades and production costs rise, the ASEAN, which occupies the dominant position in Altasia, continued to promote export-oriented economic expansion in recent years and continuously expand the scale of foreign investment by relying on population, resources, and policy advantages. Some labor-intensive manufacturing industries transfer from China to less developed Asian economies. Meanwhile, the world is undergoing the *Fifth International*

Industrial Transfer. The United States tried to rely on Asia-Pacific and Indo-Pacific regions to implement a Friend-Shoring strategy to reshape the "De-Sinicized" supply chain, which resulted in a passive industrial transfer process for China. However, to cope with the blockade and oppression of the United States, China is also taking the initiative to transfer and export industries to Southeast Asian economies. Data shows that in recent years, although China's share of imports in the United States has declined, its share in the added value of consumer goods in the United States has increased because imports from emerging markets often include components from China. Taking laptops as an example, from 2017 to 2022, the number of laptops imported by the United States from Vietnam surged. At the same time, the number of laptop components imported by Vietnam from China also increased significantly.

To highlight the significance of Altasia and ASEAN in the restructuring of *Global Value Chain (GVC)*, this paper systematically reviews the economic cooperation agreements in the Asia-Pacific and Indo-Pacific regions, as shown in Fig. 1. Clearly, Asian economies have become strategic partners actively sought after by China, the United States, and Japan, around which multilateral trade agreements and economic cooperation frameworks such as *Regional Comprehensive Economic Partnership (RCEP)*, *Indo-Pacific Economic Framework for Prosperity (IPEF)*, and *Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP)* have emerged. With the trend of regional economic integration, Asian economies are also exploring and forming new internal cooperation mechanisms to gain more significant benefits in competing with major powers.

Advantages of Altasia. While it's true that no single economy within this group can replace China, Altasia, as a collective, stands as a robust and competitive alternative. For global

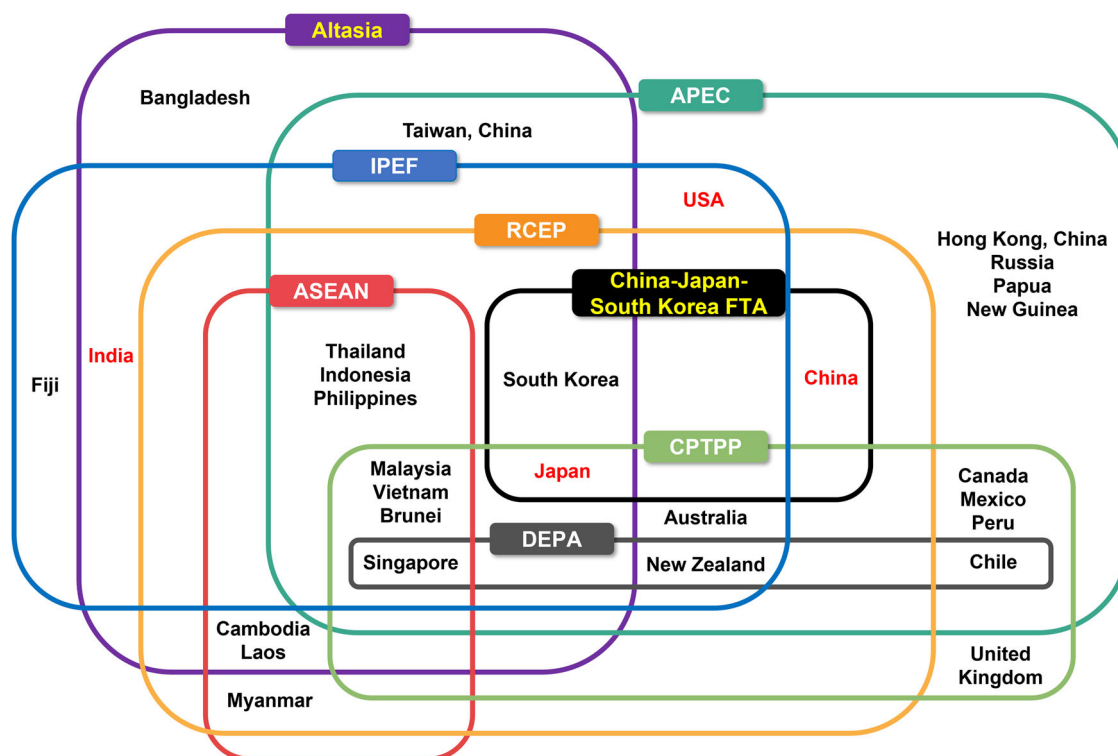


Fig. 1 Economic partnership agreement and plans in the Asia-Pacific and Indo-Pacific regions. Altasia is merely a hypothesis, and China-Japan-South Korea FTA has not been achieved, so we differentiate them by yellow font. To emphasize the major contributors to the global economy, this paper marks the names of the countries ranked in the top five of global GDP in red font.



Fig. 2 Overall advantages of Altasia compared to China in 2023. Population data are from World Population Review (retrieved from <https://worldpopulationreview.com/>); GDP, land, export and import data are from World Bank Open Data (retrieved from <https://data.worldbank.org/>); Foreign reserve data are from World Investment Report (retrieved from <https://unctad.org/>).

manufacturers seeking new production bases outside China due to geopolitical risks with the United States, Altasia offers a stable and reliable option, backed by its diverse industrial structure and a demographic dividend that is set to expand.

Altasia's strength lies in its diverse internal industrial structure. Each member economy brings a unique advantage to the table. Japan, South Korea, and Taiwan excel in technology; Singapore boasts robust logistics; Indonesia and Malaysia are resource-rich; Vietnam, Thailand, and India are developing investment policies; and the Philippines, Bangladesh, Myanmar, Laos, and Cambodia offer significantly lower labor costs compared to China. In the post-pandemic era, transnational enterprises are increasingly diversifying their industrial and supply chains to enhance risk resilience and profitability. Altasia with its array of strengths, is undoubtedly a compelling choice.

Altasia's demographic dividend is even more advantageous. Its workforce is about 1.5 times larger than China's, with 10 million more college-educated people. Taking Southeast Asian economies as the representative, compared with the rapidly aging China, Indonesia, Malaysia, Vietnam, and the Philippines have a younger population structure that will expand for a considerable period. While the labor cost is Altasia's most advantageous aspect, in terms of labor efficiency, China's average workweek is 45 h, higher than all Southeast Asian economies, and the average labor performance is 2.3 times higher than that of Southeast Asian economies. By comparing the leading macroeconomic indicators of Altasia and China (as shown in Fig. 2), we can find that Altasia's comprehensive capabilities and growth potential are comparable to China's (Hong et al. 2023).

In summary, China possesses advantages that Altasia economies cannot surpass in the short term. First, China boasts a vast economic

scale with a solid manufacturing base and extensive domestic market, enabling it to regain its significant position in the global economy post-pandemic gradually. Second, China has demonstrated considerable strength in technological innovation and infrastructure development, particularly in high-speed rail and electronic payments. Third, China's geographical proximity to the Altasia economies facilitates bilateral or multilateral trade, reducing logistics costs and accelerating the circulation of goods and services. Fourth, deep cultural connections and historical ties between China and these countries play a bridging role in promoting mutual cultural exchanges and economic cooperation. This sense of cultural closeness also helps establish and maintain trust-based business relationships.

Disadvantages of Altasia. Altasia currently holds certain comparative advantages over China in labor and manufacturing costs, which provides a solid foundation for its competitiveness. While it may not be able to close the gap with China in the short term, these advantages should not be overlooked.

As shown in Fig. 3, it is evident that Altasia economies exhibit distinct disadvantages in comparison to China across six key metrics. These disadvantages are apparent in areas such as:

- Countries like Japan, South Korea, Vietnam, Malaysia, and Indonesia have significantly lower exports of goods and services, suggesting less global trade influence than China.
- Although South Korea and Vietnam have substantial high-tech exports, other nations like Indonesia lag considerably behind, indicating a gap in high-tech industry capacity.
- Japan is the only country with total reserves exceeding one trillion USD, while other economies like Malaysia and

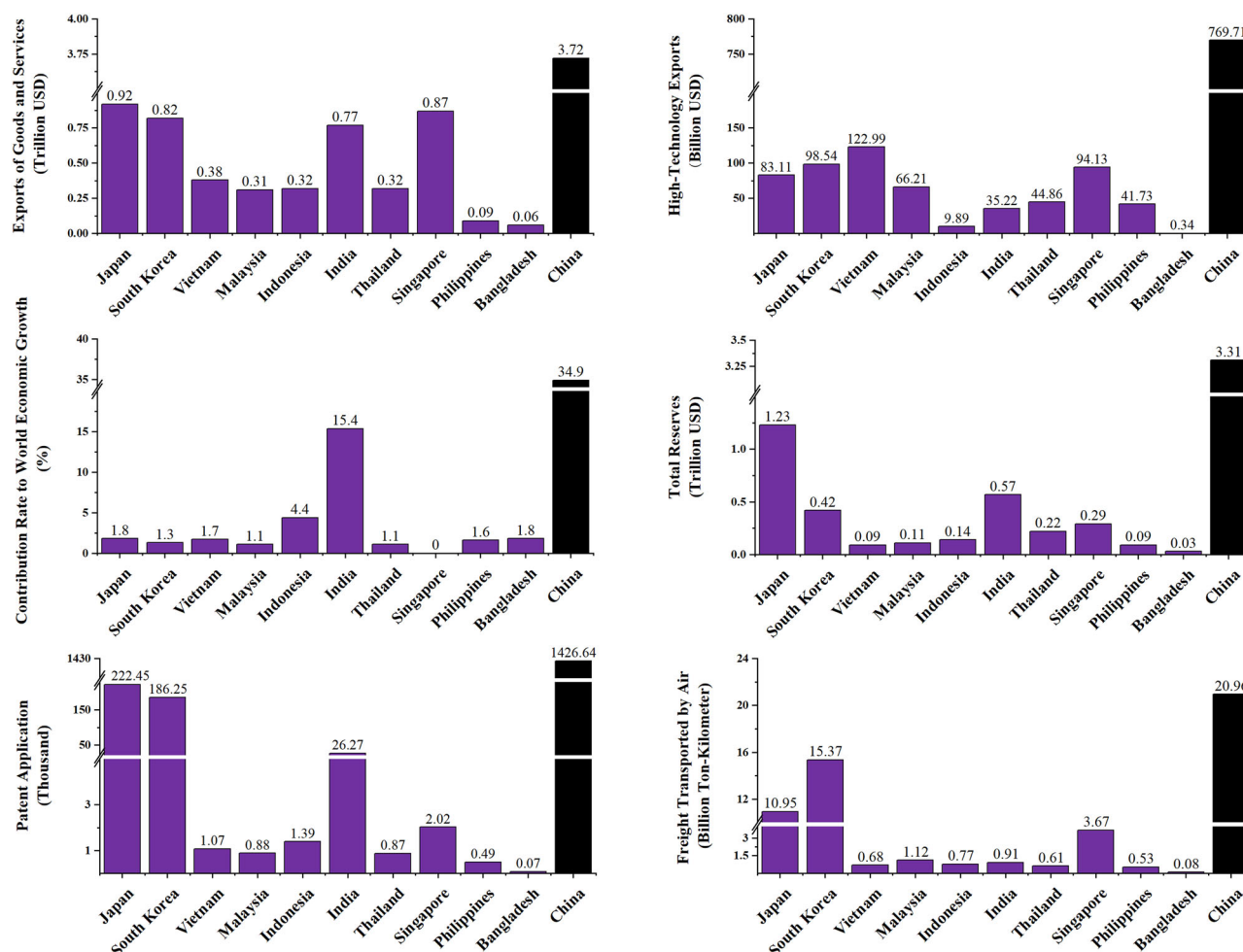


Fig. 3 Disadvantages of Altasia economies compared to China in 2023. Export of goods and services, high-technology exports, and total reserves data are from World Bank Open Data (retrieved from <https://data.worldbank.org/>); Contribution rate to world economic growth data is from World Economic Outlook Databases (retrieved from <https://www.imf.org/>); Patent application data are from International Patent System (retrieved from <https://www.wipo.int/>); Freight transported by air data is from Global Outlook for Air Transport (retrieved from <https://www.iata.org/>).

Vietnam have much smaller reserves, highlighting a disadvantage in economic buffering capacity.

- While Indonesia shows a high contribution rate of 4.4%, other countries like Malaysia (1.1%) and South Korea (1.3%) contribute less to global economic growth, which may reflect lesser economic dynamics or influence.
- There is a drastic difference in the number of patent applications, with Japan and South Korea filing hundreds of thousands, whereas countries like Vietnam and Malaysia file significantly fewer, suggesting lower innovation output.
- The volume of air freight measured in billion ton-kilometers also shows disparities; Japan and South Korea manage more substantial air freight operations, whereas nations like Vietnam and Indonesia handle much less, indicating less developed logistics and air transport capabilities.

These metrics collectively indicate that despite their strengths, these Asia-Pacific economies face considerable disadvantages in these areas compared to China, affecting their competitive stance and economic resilience on a global scale. Therefore, without external forces driving Altasia to become a tightly integrated economic entity, it could not pose a threat to China in a short period. However, an external force has emerged: the U.S. trade policy towards China.

Importantly, all Altasia economies have forged close economic ties with China, which is their leading trading partner as detailed

in Fig. 4. China is not only a source of raw materials and components but also a significant market for final products. This interdependence, a result of decades of economic integration and specialization of regional production networks, suggests that diversifying parts of the supply chain away from China may be a goal for specific industries or companies. Maintaining diplomatic and economic relations with China is necessary for regional peace and prosperity.

Literature review

Although Altasia is a fictional economy, its substitution effect on China's industrial and supply chains is becoming increasingly apparent in the context of *Global Value Chain (GVC)* restructuring. Essentially, this substitution effect represents an international industrial transfer from China to Altasia, driven by changes in the comparative advantages of these emerging economies and the United States' trade policy towards China. This section provides an overview of the new world economic order and introduces what we can do with network science to study this new economic phenomenon.

Industrial transfer in the process of globalization and regionalization. *Industrial Transfer*, an economic process in which enterprises in one country or region shift specific industries'

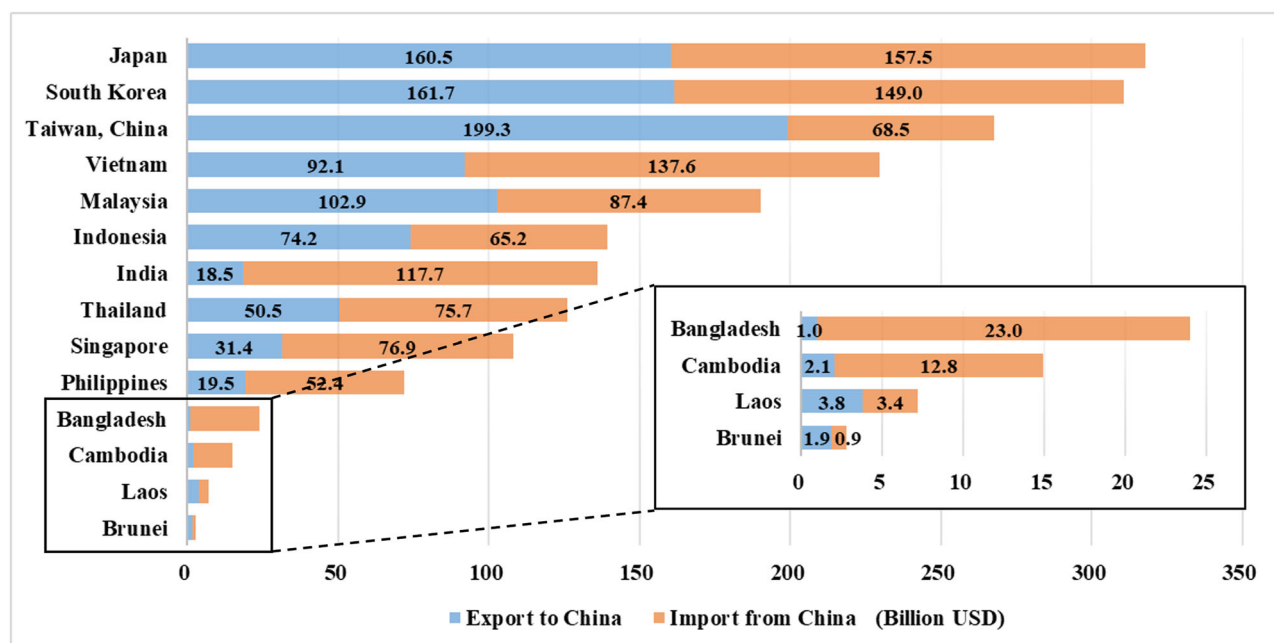


Fig. 4 China's trade with Altasia economies in 2023. The relevant data are from UN Comtrade Database (retrieved from <https://comtradeplus.un.org/>).

production, sales, or even R&D to others (Shafik and Bandyopadhyay 1992; Tian et al. 2019), is often accompanied by the transfer of market, technology, and cross-regional investment. Driven by the market mechanism, production factors will flow to sectors with high productivity or high growth rates, and the resulting *Structural Dividend* and *Technology Spillover* significantly impact the productivity of enterprises in various economies (Timmer and Szirmai 2000; Yue et al. 2022), driving the economic development of the whole country or region. The process of industrialization has brought significant changes globally, promoting the industrial transfer from developed economies to less developed ones. Since the reform and opening up, China has played a pivotal role, undertaking the large-scale international industrial transfer of developed economies. Its comparative advantage, rooted in various factors (Jiang et al. 2018), has made China the biggest beneficiary of the *Fourth International Industrial Transfer*, underscoring its global economic influence.

Traditional theories, such as the Flying Geese Paradigm (Akamatsu 1937), Core and Periphery Theory (Prebisch 1962), Product Life Cycle Theory (Vernon 1966), Marginal Industry Expansion Theory, Labor-Intensive Industry Transfer Theory (Lewis 2015) and New Economic Geography Theory (Krugman 1991) and so on, mainly involve the vertical and horizontal international industrial transfer under the background of globalization, but not the current regionalization. As the scale of GVC restructuring expands in depth and breadth, the academic community introduced the concept of the GVC (Gereffi et al. 2005), as well as developed the *GVC Accounting System* (also known as *Value-Added Trade Accounting System*) to study the processes of value creation and distribution in globalized production. This analytical framework represents a significant theoretical extension of *Input-Output Analysis (IOA)*. Since the twenty-first century, the compilation and public release of *Multi-Regional Input-Output (MRIO)* tables have enriched the theoretical framework of IOA and pioneered a new research paradigm for GVC accounting. With the continuous updates and improvements of the MRIO databases, using MRIO tables to account for the value of industrial transfer has become an essential method for studying this phenomenon from a macro perspective in recent years.

MRIO tables contain rich information on inter-regional industrial and supply chain linkages, providing a basis for assessing the direction and intensity of industrial transfer in the GVC. For example, Liu et al. (2011) and Gao et al. (2018) proposed methods for calculating the value of inter-regional industrial transfer based on the IOA framework; Gao et al. (2022) introduced a framework for measuring the scale of global industrial transfer from a production perspective; Yin et al. (2024) developed a model for measuring international industrial transfer, distinguishing between domestic and foreign-funded enterprises. Empirical research focusing on this aspect is relatively scarce, and the MRIO databases exhibit considerable lag, forcing most studies to infer future trends from past economic phenomena.

Friend-shoring strategy's impact on industrial transfer among Asian economies. In recent years, the trade policy implemented by the United States towards China has disrupted the normal patterns of international industrial transfer. On the one hand, the United States has committed to *"De-Sinicization"* to ensure the security of its critical industrial and supply chains; on the other hand, this Cold War mentality also aims to curb the rapid development of China's economy. Since 2021, the Biden administration has vigorously promoted its strategy of *"Friend-Shoring"* and *"Near-Shoring"* to strengthen supply chains and reduce reliance on geopolitical rivals. This enforced a gradual shift of production capacity from some China-based, manufacturing-centric enterprises to Southeast Asian and Latin American economies, where production costs are more advantageous. In this context, *Friend-Shoring Strategy* refers to the United States shifting its industrial and supply chains from countries like China and Russia to political allies such as ASEAN, East Asia, and South Asia, centered around its national security strategy. Implementing this policy could lead to the exclusion of China from some crucial strategic industrial sectors and cause China to become disconnected and marginalized in the GVC. Given Altasia's unique geographical location, it could become a force under the influence of European and American countries to contain China, adversely affecting the security of China's industrial and supply chains.

From an academic perspective, research on the impact of friend-shoring strategy remains largely theoretical. Boeckelmann et al. (2023) emphasize the conditions for the emergence of the concept of friend-shoring from a GVC perspective, addressing major economic and environmental issues and the limitations of its implementation. Banaszyk (2023) elaborates on how reshoring and friend-shoring are significant factors in changing the geography of international industrial and supply chains. On the empirical side, Vivoda and Matthews (2023) have indicated that friend-shoring is a practical approach to resolving the issues of industrial and supply chains for essential mineral products in the West.

In summary, current studies on the impact of the United States' friend-shoring strategy and Altasia on China are limited. Most of them rely on international trade data from the United Nations Commodity Trade Statistics Database (UN Comtrade), making them difficult to distinguish the effects of other exogenous variables, such as the COVID-19 pandemic and the Russia-Ukraine conflict. Most critically, these studies often can only infer which industrial sectors in China might be affected through import and export data, without precisely identifying whether the trend of regional economic integration in Asia, guided by U.S. policies, has led to the relocation of China's industrial and supply chains.

Network science's contribution to the study of industrial transfer. *Network Science* is a meaningful way to represent international trade flows (De Benedictis and Tajoli 2011; Fronczak and Fronczak 2012; Nemeth and Smith 1985; Smith and White 1992). Its essential advantage is in capturing the complex structure of interactions between a large number of economies and their industrial sectors. This method cannot only identify the characteristics of trade networks but also provide a comprehensive global perspective, offering solutions to various problems in observation, modeling, and forecasting (Fagiolo et al. 2013; Medo et al. 2018; Saracco et al. 2015; Schweitzer et al. 2009; Squartini et al. 2018; Chaney 2014; Ren et al. 2020). In recent years, network science and *World Economics* have been deeply integrated, resulting in significant progress in the *GVC Network Accounting System* (Xing et al. 2016, 2017a, b, 2018a, b, 2019, 2020, 2021a, b; Xing and Han 2022). This system not only reflects the global economic system's complexity, topological structure, hierarchy, and synergy but also analyzes the essence of various socio-economic phenomena from the *Systems Theory* perspective. As its most important module, the *Global Production Network (GPN)* describes the trade of intermediate goods with a high degree of generalization and restoration of the global industrial and supply chains. It has the dual characteristics of structural complexity and evolutionary complexity. Therefore, studying industrial transfer from a complex network perspective can better depict and analyze the complex real-world economic issues.

The intrinsic motivations of industrial transfer can be explained through the *Local-World Evolving Network (LWEN)* model (Li and Chen 2003). Barabási and Albert proposed the BA model in 1999, which is a simple fusion of two mechanisms: growth and preferential attachment, and it explains the evolutionary mechanism of the topological structure of complex systems in the real-world (Barabási and Albert 1999). However, the evolutionary complexity of some systems is not only derived from their structural complexity but also related to certain characteristics of the evolving entities themselves. For example, in the global economic system, primarily carried by the GPN (Ernst and Kim 2001), *Global Trade Network (GTN)* (Liu et al. 2021), and *Global Financial Network (GFN)* (Coe et al. 2014), various economic ties between countries (regions) and their industrial sectors are not established at the global level according to the

preferential attachment mechanism but tend to develop selectively within multilateral trade agreements or regional economic cooperation organizations where information is relatively transparent and risks are relatively controllable. The construction algorithm of the LWEN model (Wang et al. 2012):

- **Growth:** the network initiates with m_0 nodes and e_0 edges. With each addition of a new node to the network, it introduces m additional edges that connect the new node to existing nodes.
- **Local-World Preferential Attachment:** to simulate the local-world phenomenon, for each new node added, a subset of M existing nodes $M \geq m$ is randomly selected from the network to form a "local world". The new node then connects to m nodes within this local world. The probability of connecting to a v_i within the local world is determined by the preferential attachment probability $\prod_{local} K(i)$, defined as:

$$\prod_{local} K(i) = \frac{\prod_{i \in LW} K(i)}{\sum_{j \in Local} K(j)} \equiv \frac{M}{m_0 + t} \frac{K(i)}{\sum_{j \in Local} K(j)} \quad (1)$$

where $K(i)$ and $K(j)$ represent the degree of v_i and v_j respectively, and the summation in the denominator extends over all nodes within the local world.

This formula captures the essence of preferential attachment within a localized subset of the network by mimicking how real-world networks grow and evolve. At each moment, the new node selects m nodes to connect with from its local world according to the preferential attachment principle, rather than choosing from the entire network as in the BA model. The rule for constructing a node's local world is determined by the actual situation—For the global economic system, the multilateral trade agreements signed by a nation or the economic cooperation organizations it belongs to can be used as the basis for division, such as the RCEP, IPEF, CPTPP, APEC, ASEAN, DEPA and so on. In summary, building the GPN is one of the best ways to analyze the issue of industrial transfer on a global scale.

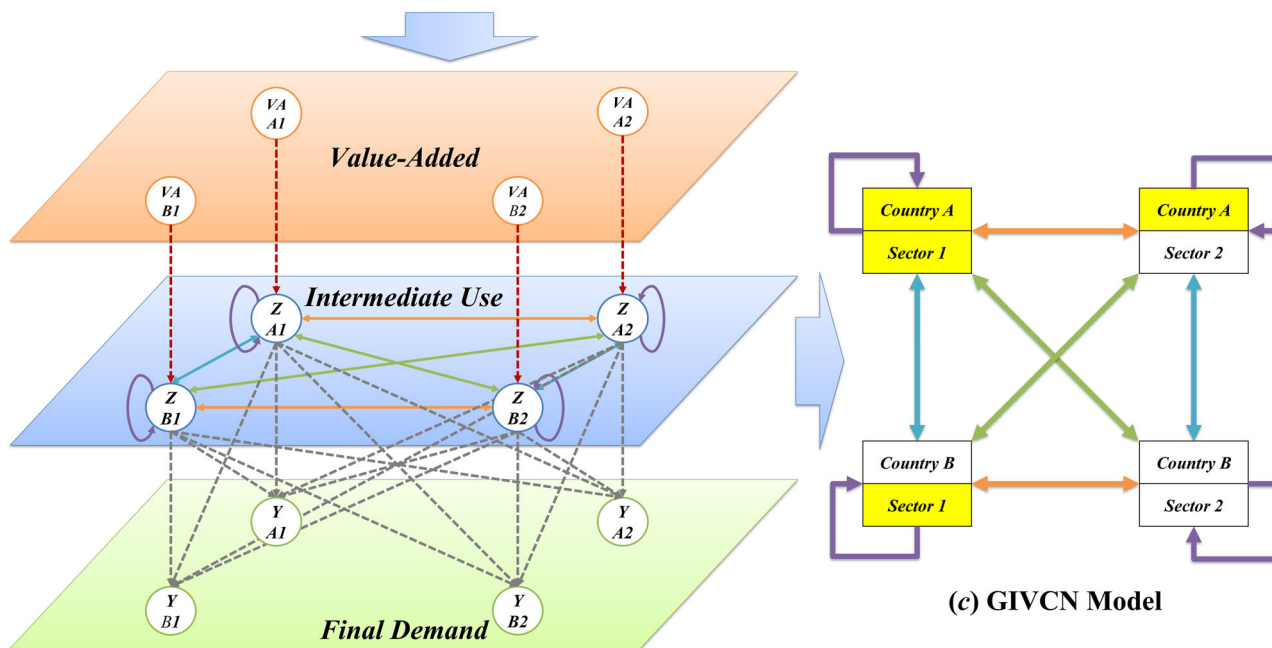
Methodology

Data structure of MRIO table. In a global economic environment increasingly characterized by fragmented and internationally distributed production processes and by trade in intermediates, the necessity for pertinent quantitative information of sufficient granularity is more urgent than ever. As mentioned above, MRIO tables quantitatively depict international trade among multiple regions and the corresponding flows of intermediate and final goods, reflecting the production technology connections and supply-demand balance relationships among various industrial sectors within economies. Also, MRIO tables can be extended according to scholars' research focuses. In recent years, various international organizations and academic institutions have compiled and released new MRIO tables, further promoting the development of GVC accounting in application fields. At the same time, as MRIO tables vividly illustrate the intricate production-consumption relationships among various economies and their industrial sectors in a chessboard-like balance sheet pattern, they are conducive to combining network science models and algorithms to analyze GVC and provide powerful tools for researching vertical specialization, international trade, and industrial structure optimization.

As shown in Fig. 5a, the final demand and intermediate use sections of the global MRIO table constitute the complex topological structures of the GTN and GPN, respectively. The MRIO table is the most comprehensive data structure for studying the efficiency of distribution and production in the global economic system. It explains which consumers use the various resources input into the production system and how

Two Countries with Two Sectors		Intermediate Use (Z)				Final Demand (Y)			
		Country A		Country B		Country A		Country B	
		Sector 1	Sector 2	Sector 1	Sector 2	Sector 1	Sector 2	Sector 1	Sector 2
Country A	Sector 1								
	Sector 2								
Country B	Sector 1								
	Sector 2								
Value-Added (VA)									

(a) MRIO Table Including Two Countries with Two Sectors



(b) Multi-Layer Structure of MRIO Table

(c) GIVCN Model

Fig. 5 Relationship between MRIO table and network. **a** An example of the MRIO table including two countries with two sectors; **b** its multi-layer structure; **c** intermediate goods flowing among countries and sectors. The typical MRIO table includes three areas: value-added, intermediate use, and final demand. The whole global economic system can be extracted into a multi-layer network, as shown in **(b)**, which includes three layers: the value-added layer, the intermediate use layer, and the final demand layer. The intermediate use layer depicts the topological structure of GPN, in which the nodes are industrial sectors, and the edges are the intermediate goods from the upstream sectors to the downstream ones.

much they consume. Additionally, they can link regional and sectoral outputs with satellite accounts such as energy use, carbon emissions, and environmental pollutants, providing a data foundation for extensive research on environmental impacts. Economists have developed a mature GVC accounting system using the global MRIO table (Hummels et al. 2001; Koopman et al. 2010; Timmer et al. 2012; Wang et al. 2017; Mi et al. 2018).

Currently, representative world MRIO databases include the World Input-Output Database (WIOD), the OECD-WTO Database on Trade in Value-Added (OECD-TiVA), Analytical Multinational Enterprises and Global Value Chains Database (OECD-AMNE); the Eora Multi-Region Input-Output Table Database (Eora-MRIO), the Asian Development Bank (ADB) Multi-Regional Input-Output Tables (ADB-MRIO) and China Emission Accounts and Datasets (CEADs). The basic information of these MRIO databases is shown in Table 1.

From the perspective of data structure: the WIOD, OECD-TiVA (Timmer et al. 2015), Eora-MRIO (Lenzen et al. 2013),

ADB-MRIO, and EMERGINGV2 in the CEADs databases are all Inter-Country Input-Output (ICIO) tables, which can be used to study the transfer of industries from all enterprises in a country/region to foreign industries; the CHNMRI and CHNCLMRI in the CEADs database are Inter-Region Input-Output (IRIO) tables, which can expand specific countries or regions in the ICIO tables to obtain Regionally Extended Multi-Region Input-Output (RE-MRIO) tables, thereby considering the trend of industrial transfer within countries or regions; the OECD-AMNE database is a Multi-Regional Multiple-Input Multiple-Output (MRMIMO) table, which can further distinguish between domestic and foreign-funded enterprises based on the ICIO tables.

To facilitate analysis work related to the Asia and the Pacific Region, the ADB has produced MRIO tables building on the WIOD database (Timmer et al. 2015) to include 29 Asian economies, namely: Armenia, Australia, Bangladesh, Bhutan, Brunei Darussalam, Cambodia, Fiji, Georgia, Hong Kong, China, India, Indonesia, Kazakhstan, Kyrgyz Republic, Lao People's Democratic Republic,

Table 1 Major MRIO databases.					
Database	Version	Time span	Country/ Region	Industrial sector	Abbr.
WIOD	Long-Run	1965–2000	26	35	WIODLR
	2016 Release	2000–2014	44	56	WIOD2016
	2013 Release	1995–2011	41	35	WIOD2013
OECD-TiVA	2022 Release	1995–2020	77	45	TiVA2022
	2021 Release	1995–2018	68	45	TiVA2021
	2018 Release	2005–2015	65	36	TiVA2018
	2016 Release	1995–2011	64	34	TiVA2016
	2015 Release	1995, 2000, 2005, 2008–2011	62	34	TiVA2015
OECD-AMNE	2023 Release	2000–2019	77	41 × 2 = 82	AMNE2023
	2018 Release	2005–2016	60	34 × 2 = 68	AMNE2018
Eora-MRIO	2023 Update V199.82	1990–2016	189	26	Eora26
ADB-MRIO	2023 Update, 62 economies	2000, 2007–2022	63	35	ADB2023 (E62)
	2023 Update, 71 economies (including Latin America)	2007, 2011, 2017	72	35	ADB2023 (E71)
CEADs	2023 Update, 72 economies	2017–2022	73	35	ADB2023 (E72)
	EMERGING Model V2	2010, 2015–2019	245	135	EMERGINGV2
	China MRIO Table	2012, 2015, 2017	31	42	CHNMRI0
	China City-Level MRIO Table	2012, 2015, 2017	313	42	CHNCLMRI0

Table 2 Countries or regions' names and abbreviations in ADB.			
Abbr.	Country/region	Abbr.	Country/region
AUS	Australia	NOR	Norway
AUT	Austria	POL	Poland
BEL	Belgium	POR	Portugal
BGR	Bulgaria	ROM	Romania
BRA	Brazil	RUS	Russia
CAN	Canada	SVK	Slovak Republic
SWI	Switzerland	SVN	Slovenia
PRC	People's Republic of China	SWE	Sweden
CYP	Cyprus	TUR	Turkey
CZE	Czech Republic	TAP	Taipei, China
GER	Germany	USA	United States
DEN	Denmark	BAN	Bangladesh
SPA	Spain	MAL	Malaysia
EST	Estonia	PHI	Philippines
FIN	Finland	THE	Thailand
FRA	France	VIEW	Vietnam
UKG	United Kingdom	KAZ	Kazakhstan
GRC	Greece	MON	Mongolia
HRV	Croatia	SRI	Sri Lanka
HUN	Hungary	PAK	Pakistan
INO	Indonesia	FIJ	Fiji
IND	India	LAO	Lao People's Democratic Republic
IRE	Ireland	BRU	Brunei Darussalam
ITA	Italy	BHU	Bhutan
JPN	Japan	KGZ	Kyrgyz Republic
KOR	Republic of Korea	CAM	Cambodia
LTU	Lithuania	MLD	Maldives
LUX	Luxembourg	NEP	Nepal
LVA	Latvia	SIN	Singapore
MEX	Mexico	HKG	Hong Kong, China
MLT	Malta	ROW	Rest of the World
NET	Netherlands		

Malaysia, Maldives, Mongolia, Nepal, New Zealand, Pakistan, People's Republic of China, Philippines, Republic of Korea, Singapore, Sri Lanka, Taipei, China, Thailand, and Vietnam. As a result, ADB compiles MRIO tables at current prices, and at constant

2010 prices for 62 economies and aggregated Rest of the World (RoW). This has facilitated the production and analysis of global value chain related statistics for Asian economies. Economies explicitly identified in the ADB-MRIO account for at least 93% of the world Gross Domestic Product.

This paper focuses on the impact of Asian economies on the *China Production Network (CPN)*, and requires high timeliness of data. Therefore, considering the characteristics of the above MRIO databases, this paper ultimately adopted the intermediate uses section of ADB2023(E62) as the modeling data. The covered countries or regions and industrial sectors are detailed in Tables 2 and 3.

Network modeling and extracting. This paper views the industrial sectors of various countries or regions as nodes, and their IO relationships via industrial and supply chains as edges, with the trade volume of intermediate goods reflecting the strength of these relationships as weights. When the total number of nodes in the network is N , the node set is represented as $V = \{v_1, v_2, \dots, v_N\}$, the edge set as $E = \{e_{11}, e_{12}, \dots, e_{ij}, \dots, e_{N(N-1)}, e_{NN}\}$, and the weight set as $W = \{w_{11}, w_{12}, \dots, w_{ij}, \dots, w_{N(N-1)}, w_{NN}\}$. While the GPN applies the weight set W to display the adjacency matrix, each row refers to the distribution of intermediate goods output from an upstream sector to several downstream sectors, and each column the intermediate goods input obtained by a downstream sector from several upstream sectors. Since the graph $G = (V, E, W)$ reflects the division of labor that each industrial sector undertakes and portrays the intermediate goods trade among economies in the GPN, this paper names it the *Global Industrial Value Chain Network (GIVCN)* model, as shown in Fig. 5c.

Obviously, the GIVCN models are highly dense (approximately fully connected) weighted directed networks, so data dimensionality reduction is required to visualize the network's topological structure changes. Considering the substantial heterogeneity of intermediate goods trade between upstream and downstream sectors, this paper uses the *X-Index Filtering Algorithm (XIFA)* to extract a sub-network from the GIVCN model (Xing and Han 2022), which is named the *Global Industrial Value Chain Backbone Network (GIVCBN)* and represented as $\hat{G} = (V, \hat{E}, \hat{W})$. Therefore, this paper builds 16 GIVCBN models based on the intermediate uses section of ADB2023(E62) from

Table 3 Industrial sectors' names and abbreviations in ADB.

Abbr.	Industrial sector	Abbr.	Industrial sector
S01	Agriculture, Hunting, Forestry and Fishing	S19	Sale, Maintenance and Repair of Motor Vehicles and Motorcycles; Retail Sale of Fuel
S02	Mining and Quarrying	S20	Wholesale Trade and Commission Trade, Except of Motor Vehicles and Motorcycles
S03	Food, Beverages and Tobacco	S21	Retail Trade, Except of Motor Vehicles and Motorcycles; Repair of Household Goods
S04	Textiles and Textile Products	S22	Hotels and Restaurants
S05	Leather, Leather and Footwear	S23	Inland Transport
S06	Wood and Products of Wood and Cork	S24	Water Transport
S07	Pulp, Paper, Paper, Printing and Publishing	S25	Air Transport
S08	Coke, Refined Petroleum and Nuclear Fuel	S26	Other Supporting and Auxiliary Transport Activities; Activities of Travel Agencies
S09	Chemicals and Chemical Products	S27	Post and Telecommunications
S10	Rubber and Plastics	S28	Financial Intermediation
S11	Other Non-Metallic Mineral	S29	Real Estate Activities
S12	Basic Metals and Fabricated Metal	S30	Renting of M&Eq and Other Business Activities
S13	Machinery, Nec	S31	Public Admin and Defence; Compulsory Social Security
S14	Electrical and Optical Equipment	S32	Education
S15	Transport Equipment	S33	Health and Social Work
S16	Manufacturing, Nec; Recycling	S34	Other Community, Social and Personal Services
S17	Electricity, Gas and Water Supply	S35	Private Households with Employed Persons
S18	Construction		

2007 to 2022. Each model includes 2205 industrial sectors, distributed across 62 economies and RoW. The topological structure of the latest model is shown in Fig. 6.

Furthermore, to reflect the potential impact of Altasia and ASEAN on China's industrial and supply chains, we need to extract three types of GIVCBN models from the GIVCN model, and the extraction methods and basic settings are shown in Fig. 7:

- *GIVCBN-I Model*: firstly, the XIFA algorithm is applied to prune the GIVCN model. Secondly, the industrial sectors of Altasia are merged, treating them as a loosely integrated entity. Finally, a dichotomized network model containing 1750 nodes is obtained, as shown in Fig. 8.
- *GIVCBN-II Model*: firstly, the industrial sectors of Altasia are merged to form a tightly integrated entity in the GIVCN model. Secondly, the XIFA algorithm is applied to prune this network. Finally, the resulting GIVCBN model is dichotomized, as shown in Fig. 9.
- *GIVCBN-III Model*: firstly, the industrial sectors of nine ASEAN member countries are merged (Myanmar is not included in Altasia), that is, ASEAN member countries are regarded as a whole. Secondly, XIFA algorithm is applied to prune this network. Finally, ASEAN member countries are merged with the remaining relevant countries or regions in Altasia, as shown in Fig. 10.

The GIVCBN-II or GIVCBN-III models treat the inner industrial sectors of Altasia economies or ASEAN member countries as a closely integrated whole, which allows them to compete for more intermediate goods resources in turn. As a result, the relatively less efficient industrial and supply chains in other economies will be removed, while Altasia's industrial sectors, initially located in the "Periphery" or "Semi-Periphery" of GPN, become closer to the "Core". Compared to the GIVCBN-I model, the GIVCBN-II or GIVCBN-III model demonstrates the network clustering in Altasia and the network disintegration in the remaining economies. Therefore, by comparing the adjacency matrices of the GIVCBN-I model and the GIVCBN-II or GIVCBN-III model, we can identify which of China's industrial and supply chains are disrupted due to the substitution effect of Altasia or ASEAN, manifesting as the risk exposure of CPN.

The network topological structures of the three types of GIVCBN models are not random but formed according to specific rules, ensuring the reliability of our findings. This non-

triviality allows us to reflect the heterogeneity of each model, which can be measured by both network-level and node-level indicators. By comparing the measurement results of the real-world network (GIVCBN-I model) and the artificial networks (GIVCBN-II or GIVCBN-III model), we can confidently quantify the extent to which different modeling considerations impact the network topology structure.

Network-level measurements

Average path length. The Average Path Length (denoted as *APL*) measures the average shortest distance between all industrial sectors in the GIVCBN model, which describes the degree of separation among nodes. In network studies, the distance between nodes is generally defined as the minimum number of edges that must be traversed from v_i to v_j . A path whose length equals the distance between two nodes is called the Shortest Path, denoted by d_{ij} . If there is no such path in the network, it is denoted as $d_{ij} = \infty$. The formula for average path length is given by:

$$APL = \frac{1}{N(N-1)} \sum_{i \neq j} d_{ij} \tag{2}$$

where $APL \in [1, \infty)$. When $APL = 1$, all pairs of nodes are directly connected, forming a fully connected network.

A low *APL* value close to 1 indicates a "small-world" effect, which means industrial sectors are closely linked to each other, facilitating quick intermediate goods trades. This is often seen within specific domestic production network or that of an economy formed through regional economic integration. On the contrary, a high *APL* value significantly greater than 1 suggests that the GIVCBN model has economies or industrial sectors that are relatively isolated or require more steps to reach, which can impact the efficiency of international trade transactions.

Average clustering coefficient. The Clustering Coefficient (denoted as *C*) measures the degree to which nodes in an undirected network tend to cluster together, indicating the familiarity among nodes. In the GIVCBN model where direction is ignored, the $C(i)$ describes the close relationships between the v_i and its directly connected neighbors. Numerically, it is the ratio of the number of edges between these adjacent nodes to the maximum possible number of edges. The formula for clustering coefficient of v_i is

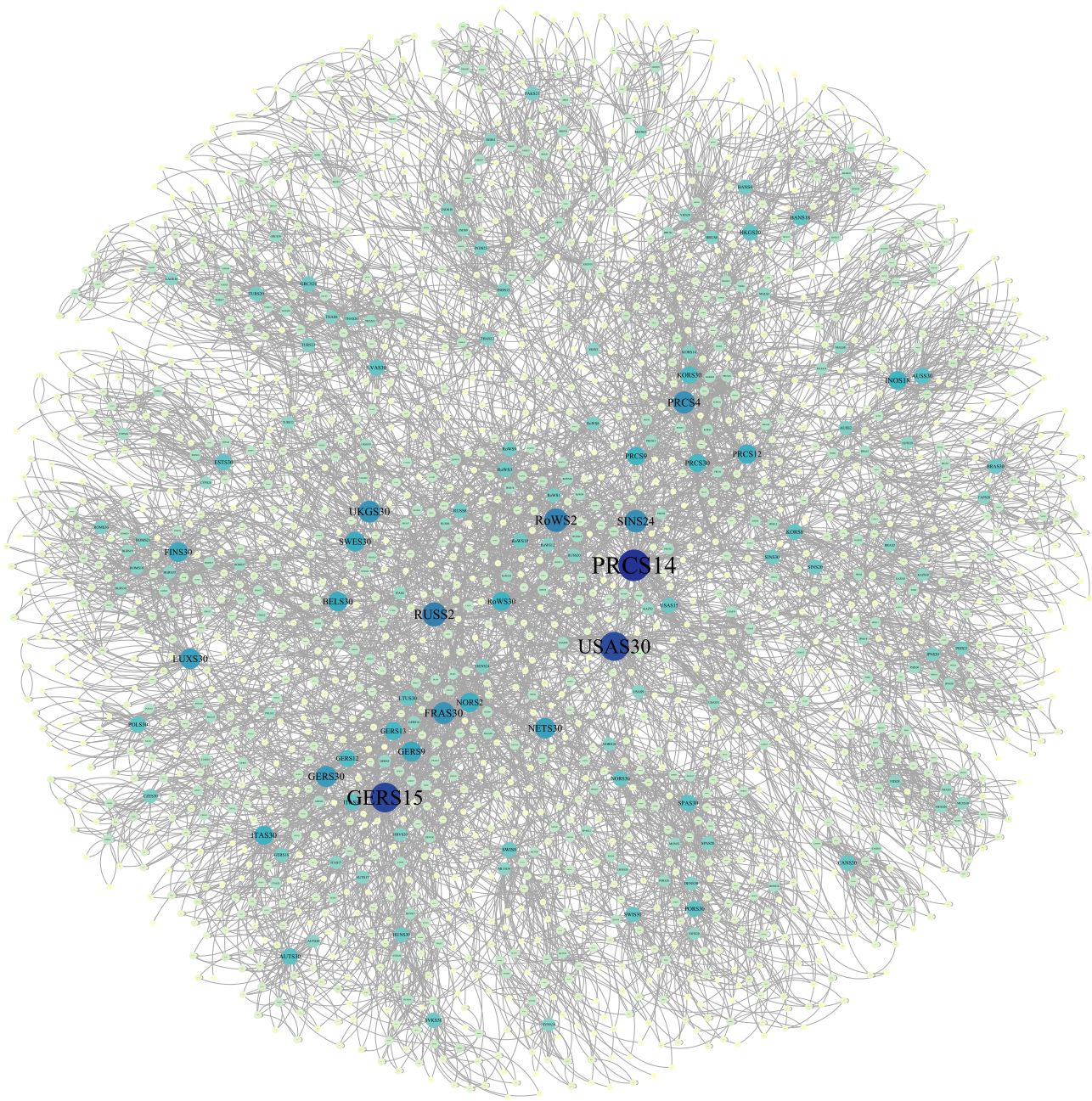


Fig. 6 Topological structure of GIVCBN-2022 model. This figure was generated by inputting the adjacency matrix of the GIVCBN-2022 model into the Gephi software for visualization.

given by:

$$C(i) = \frac{A(i)}{\frac{1}{2}K(i)(K(i) - 1)} = \frac{2A(i)}{K(i)(K(i) - 1)} \quad (3)$$

where $A(i)$ is the actual number of edges between adjacent nodes of v_i , and $K(i)$ is the degree of v_i . If v_i has only one or no neighboring node, i.e., $K(i) = 1$ or $K(i) = 0$, $A(i) = 0$, and the numerator and denominator of the formula are both 0.

The average value of these ratios across the network is known as the Average Clustering Coefficient (denoted as $\langle C \rangle$), which is equal to:

$$\langle C \rangle = \frac{\sum_{i \neq j} C(i)}{N} = \frac{1}{N} \sum_{i \neq j} \frac{2A(i)}{K(i)(K(i) - 1)} \quad (4)$$

On the one hand, a high $\langle C \rangle$ value indicates that many industrial sectors have interconnected upstream and downstream sectors, suggesting a high potential for localized intermediate goods trade relationships within the geographically adjacent regions or the same multilateral trade agreement. This phenomenon implies the formation of industrial clustering. On the other hand, a low $\langle C \rangle$ value suggests that the GIVCBN model has a more tree-like structure, with industrial sectors having fewer interconnected sectors, which might impede the spread of intermediate goods but could enhance system robustness against localized attacks.

Asymmetry. The Asymmetry (denoted as ASY) generally refers to a characteristic of directed networks where the relationships between nodes are not reciprocal. This concept is crucial in

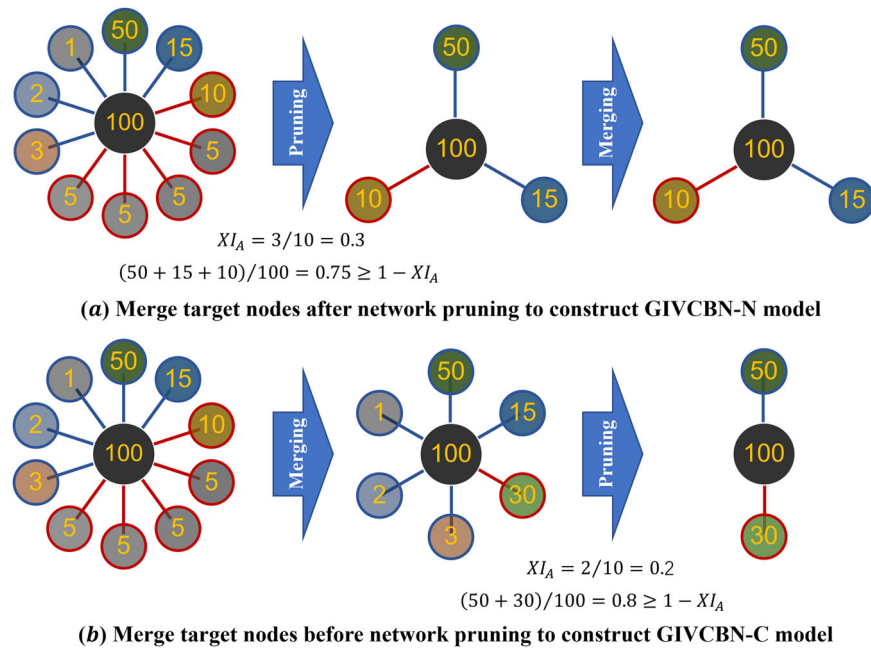


Fig. 7 Extraction steps of GIVCBN models. The XIFA algorithm is used to prune the network of an individual network (Ego Network) to extract its backbone network. Assuming that the key nodes with red outlines (assigned weights of 10, 5, 5, 5 respectively) can be merged into a super node, then in **(a)**, except for the node with a weight of 10, the other nodes have disappeared in the pruning process, so the weight assigned to the super node after merging is only 10. If, as in **(b)**, the merger is carried out first to obtain a super node with a weight of 30, then pruning will result in a completely different backbone network.

understanding the dynamics and structure of networks where the direction of interaction plays a significant role. An asymmetric relationship in the GIVCBN model where the direction is emphasized implies that: if there is an intermediate goods export from sector i in one country to sector j in another, there isn't necessarily a corresponding one from sector j to sector i . The formula for asymmetry is given by:

$$ASY = \sqrt{\sum_i (K^{IN}(i) - K^{OUT}(i))^2 / N} \quad (5)$$

where $K^{IN}(i)$ and $K^{OUT}(i)$ are the in-degree and out-degree of v_i respectively. If $ASY = 0$, the in-degree of each node is equal to the out-degree, and $ASY > 0$ proves that the in-degree and out-degree of the nodes in the network are inconsistent, and the network is asymmetric.

A higher ASY value indicates a significant disparity in the sources and destinations of intermediate goods within a given production network. This means that after an industrial sector obtains intermediate goods from N other sectors for production, the number of downstream sectors that receive its output will be significantly greater than or less than N . This phenomenon reflects a low degree of vertical specialization, which in turn negatively affects the efficiency of the production network.

Node-level measurements

Degree centrality. The Degree Centrality (denoted as DC) measures the number of direct industrial connections an industrial sector has in the GIVCBN model, intuitively reflecting its *influence scope* in the industrial and supply chains. The degree of a node is the number of edges directly connected to other nodes in the network. The greater the degree of a node, the more important it is, indicating that the node has a higher DC value. In a network with N nodes, the maximum possible degree value of a node is $N - 1$. The formula for degree centrality of v_i is

given by:

$$DC(i) = \frac{K(i)}{N - 1} \quad (6)$$

In the GIVCBN model, the higher the DC value of an industrial sector, the more sectors that establish industrial connections with it, and therefore its status in the network is relatively higher. Since the GIVCBN model is a directed network, the number of direct connections between an industrial sector and the upstream and downstream sectors on the global industrial and supply chains needs to be reflected separately by the *In-Degree Centrality* and the *Out-Degree Centrality*, represented by $DC^{IN}(i)$ and $DC^{OUT}(i)$ respectively.

Betweenness centrality. The Betweenness Centrality (denoted as BC) measures the ability of an industrial sector to act as a hub for value-added brought by intermediate goods trade in the GIVCBN model. According to Burt's theory of structural holes, it reflects the *profitability* of sectors in the industrial and supply chains. If there are d_{jk} shortest paths between a pair of nodes, and $d_{jk}(i)$ of them pass through v_i , then the contribution rate of v_i to the betweenness of this pair of nodes is $d_{jk}(i)/d_{jk}$. By summing the contribution rates of v_i to the betweenness of all pairs of nodes in the network and dividing by the total number of node pairs, we can obtain the betweenness of v_i . The formula for betweenness centrality of v_i is given by:

$$BC(i) = \sum_{i,j,k \in \{1,2,\dots,N\}} \frac{2d_{jk}(i)}{(N-1)(N-2)d_{jk}} \quad (7)$$

where d_{jk} is the number of shortest paths from v_j to v_k , and $d_{jk}(i)$ is the number of those paths that pass through v_i .

According to its concept and formula, BC can be used to quantitatively analyze the potential control ability of industrial sectors over intermediate goods trade. In other words, the greater the BC value of an industrial sector, the stronger its ability to

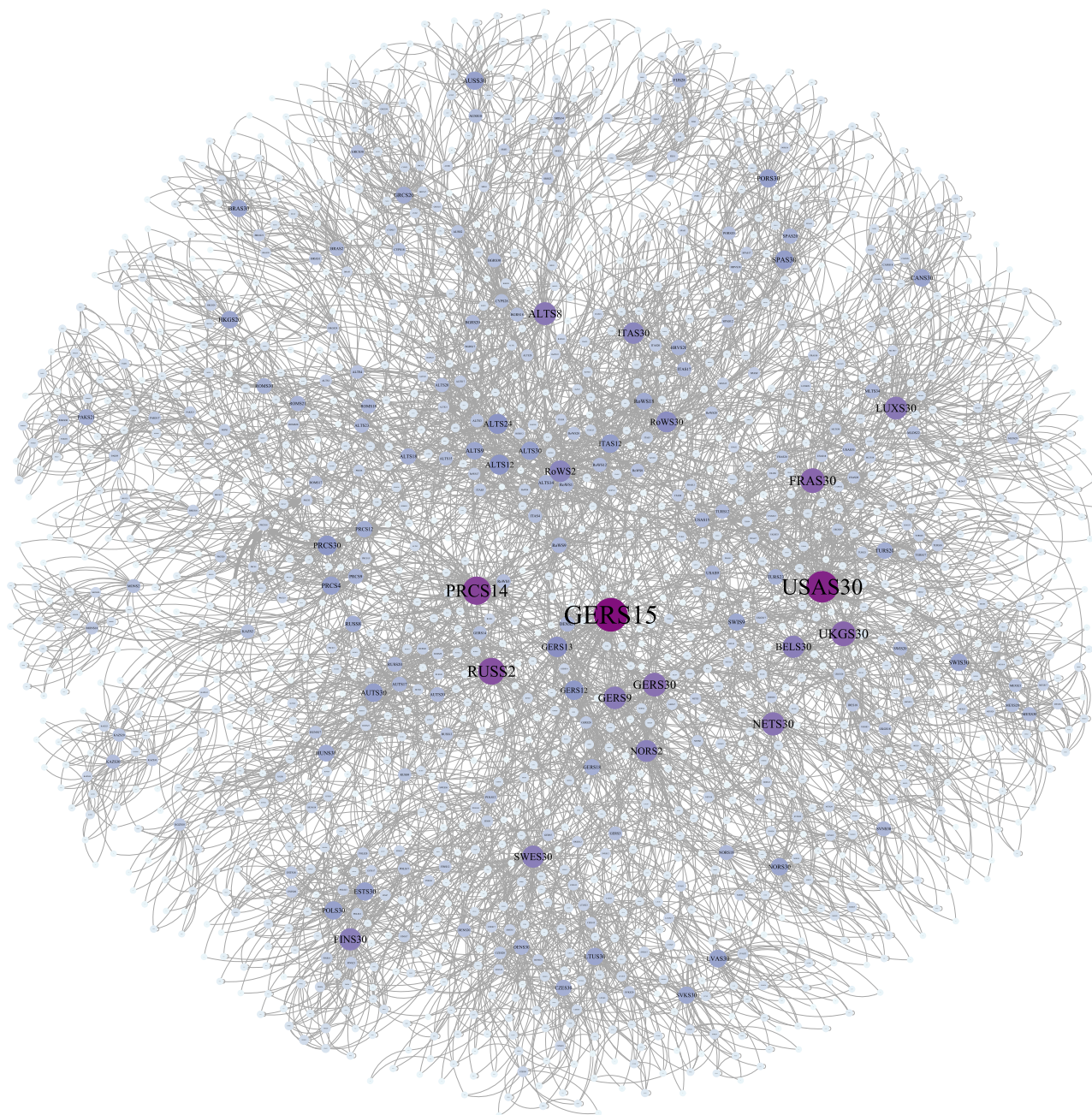


Fig. 8 GIVCBN-I-2022 model. This figure was generated by inputting the adjacency matrix of the GIVCBN-I-2022 model into the Gephi software for visualization.

control value flows and the information advantage it possesses. If an industrial sector with high *BC* value is removed or restricted, there will be a significant impact on the network topology and transmission efficiency.

Closeness centrality. The Closeness Centrality (denoted as *CC*) measures the industrial agglomeration relationship between an industrial sector and all its upstream or downstream sectors in the GIVCBN model. A higher *CC* value indicates that an industrial sector has a higher *robustness of risk*, which means it has the ability to withstand uncertainties arising from upstream or downstream sectors in the GPN. Numerically, the *CC* value is equal to the reciprocal of the sum of the shortest distances from one node to all other nodes. The formula for closeness centrality

of v_i is given by:

$$CC(i) = \frac{N - 1}{\sum_{j \neq i} d_{ij}} \quad (8)$$

where d_{ij} is the shortest distance from v_i to v_j .

When the sum of the shortest distances from v_i to all other nodes is minimized, the *CC*(*i*) value is maximized. The node with the highest *BC* has the greatest control over the flow of value in the network, while the node with the highest *CC* value has the best observational perspective on the flow of value. In the GIVCBN model, if an industrial sector is very close to other industrial sectors, it will be able to conduct intermediate goods trade efficiently and quickly, thereby greatly enhancing the robustness of its related industrial and supply chains.

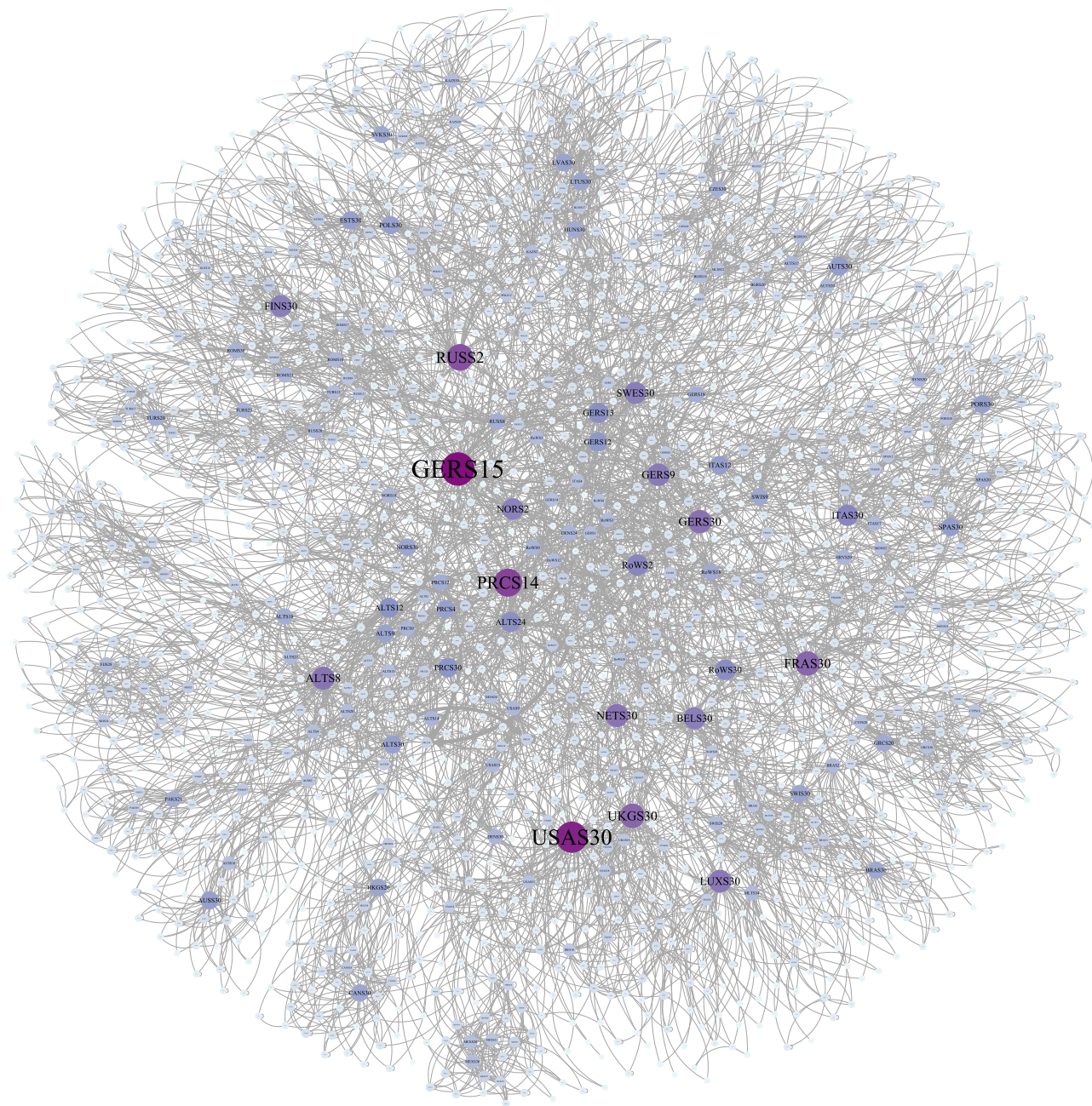


Fig. 9 GIVCBN-II-2022 model. This figure was generated by inputting the adjacency matrix of the GIVCBN-II-2022 model into the Gephi software for visualization.

Furthermore, this concept can be expanded used to assess the industrial agglomeration phenomena formed by specific industrial sectors with their upstream and downstream sectors through *In-Degree Closeness Centrality* and *Out-Degree Closeness Centrality*, represented by $CC^{IN}(i)$ and $CC^{OUT}(i)$ respectively.

Results

This section primarily employs counterfactual analysis and complex network characteristic indicators to explore the risk exposures within the CPN. It quantitatively assesses the extent to which strengthening Altasia or ASEAN into a powerful regional economic alliance could diminish China's function and status in the GPN. Additionally, it empirically analyzes the risk of industrial chain relocation occurring in several kinds of factor-intensive industries of China.

Risk exposure of China's industrial and supply chains. To sort out the risk exposure of China's industrial and supply chains from a topological structure perspective, particularly the upstream and downstream industrial sectors that might face decoupling or disruption, this paper compares all the linkages in the CPN between the counterfactual model and the null model of GIVCBN. It was found that some of these linkages were retained in the extraction process of the null model but were deleted in the counterfactual model due to their decreased relative importance. Due to space limitations, the paper uses chord diagrams to list the comparative results from 2019 to 2022, as shown in Figs. 11 and 12. Each chord in the chord diagram represents an industrial and supply chain that exists in the null model but disappears in the counterfactual model, indicating potential risk exposures within the CPN when Altasia becomes a reality. It is

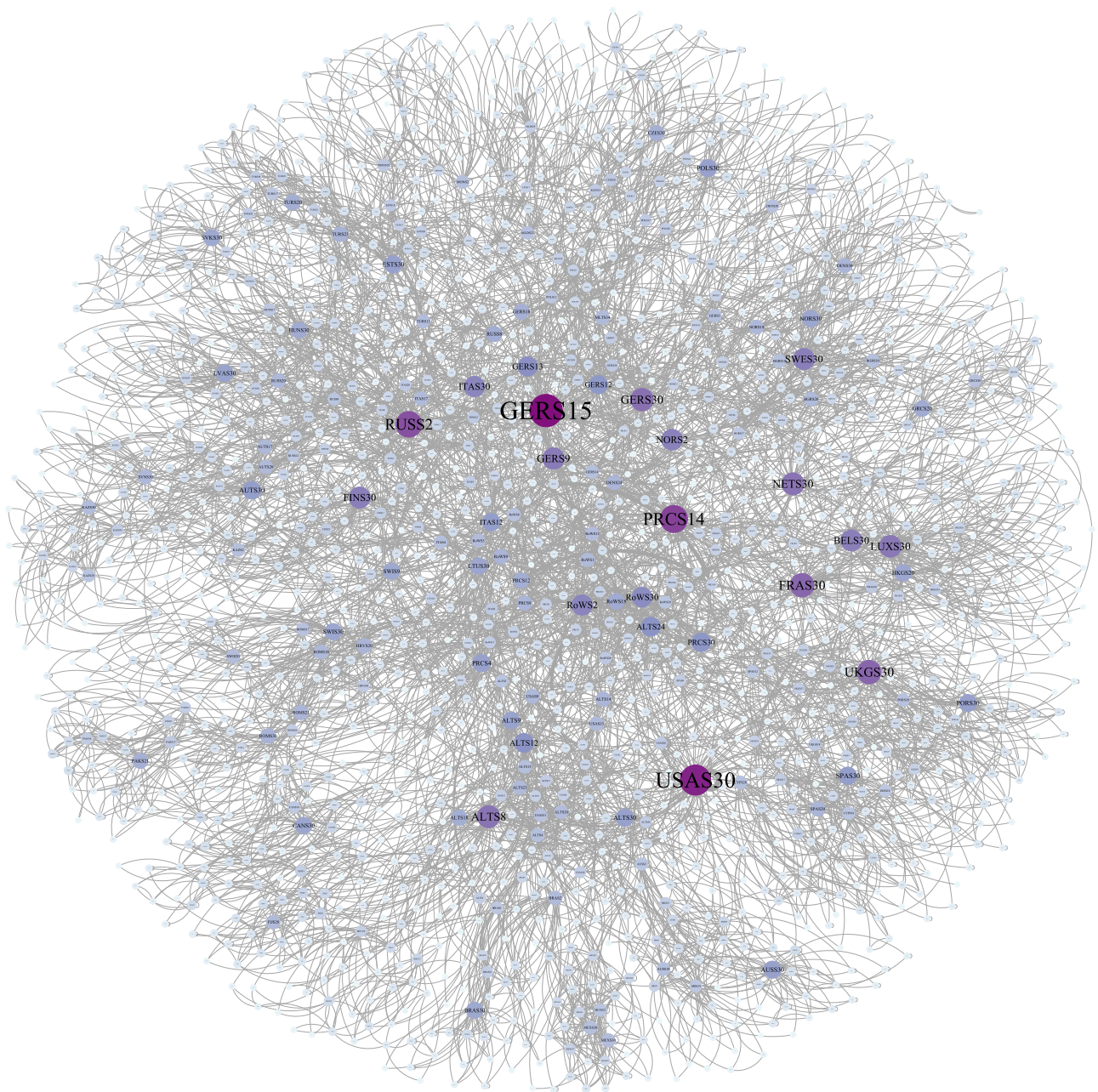


Fig. 10 GIVCBN-III-2022 model. This figure was generated by inputting the adjacency matrix of the GIVCBN-III-2022 model into the Gephi software for visualization.

important to note that the GIVCBN-III model is closer to reality than the GIVCBN-II model, in which ASEAN member countries are promoting regional trade and investment through multilateral agreements and joint market plans, such as the *ASEAN Free Trade Area (AFTA)*. These countries work together to reduce trade barriers and strengthen regional economic integration, which enhances economic interconnectivity among them and strengthens their connections with other global economies.

By comparing the GIVCBN-I model (null model) and the GIVCBN-II or GIVCBN-III model (counterfactual model), the risk exposure in CPN can be observed due to the substitution effect of the Altasia, namely the decoupling phenomena between industrial sectors. Tables 4 and 5 provide statistics on the risk exposure of CPN under two scenarios. The first column in Tables 4 and 5 lists the 35 industrial sectors as upstream sectors at risk, and each subsequent column lists the names of the

downstream sectors at risk in specific years. If, in any year, an upstream sector forms a risk exposure with at least one downstream sector, the table will list the names of these downstream sectors. In this case, the combination of upstream and downstream sectors reflects the specific industrial and supply chains where decoupling phenomena have occurred. If an upstream sector does not form a risk exposure with any downstream sector in a particular year, it is indicated with “—”.

From the results of comparing GIVCBN-I with GIVCBN-II and GIVCBN-III respectively, we can observe the following facts. First of all, considering all the Altasia-related economies as a whole will cause about 25 of China’s industrial and supply chains to be broken, while considering ASEAN as a whole will cause less than 15. We believe that the reason behind this result lies in the strong industrial complementarity between Japan, South Korea, India, and Bangladesh and the ASEAN member

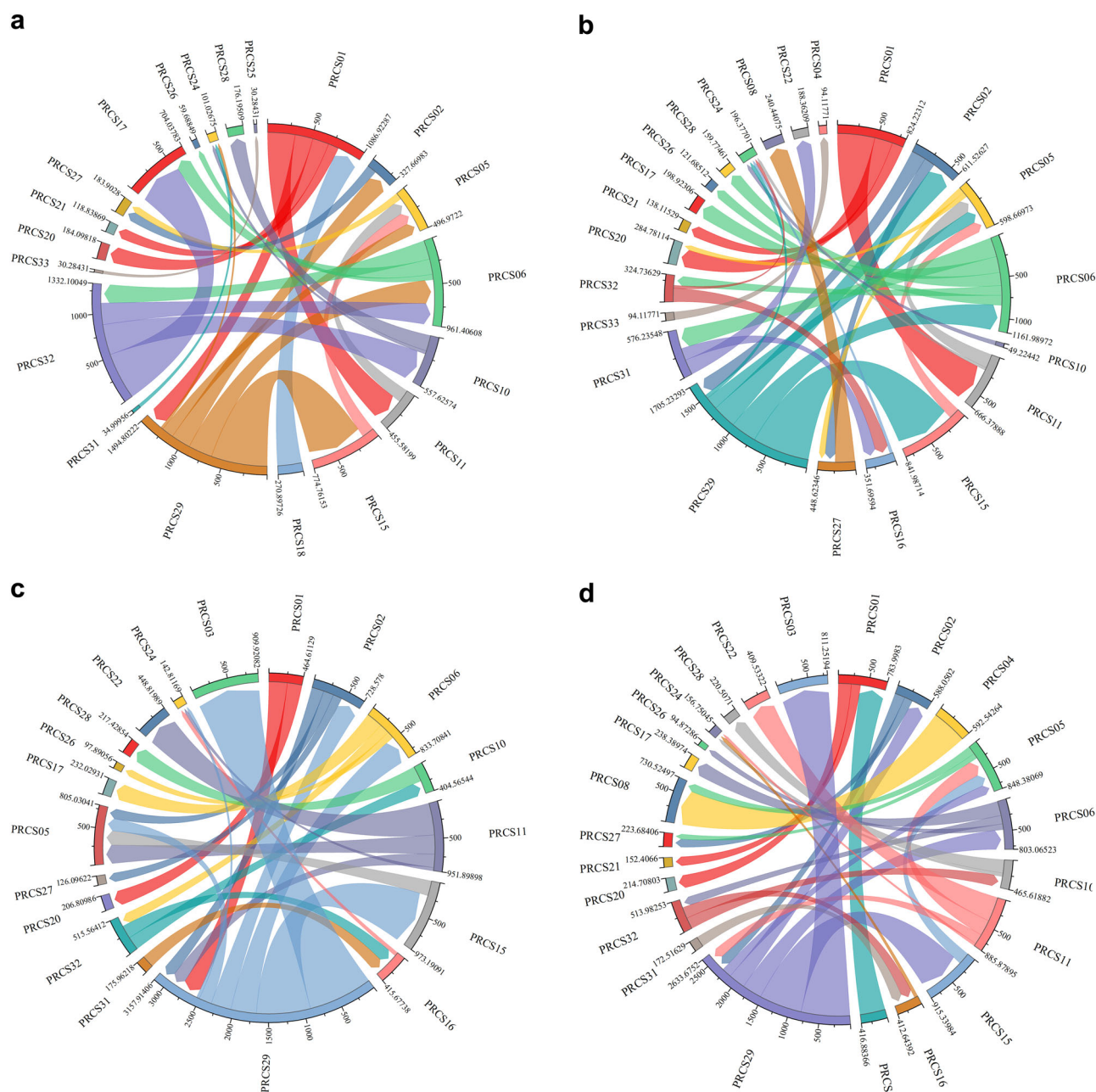


Fig. 11 Changes in the topological structure of CPN by comparing GIVCBN-I and GIVCBN-II models. The width of the chord is proportional to the intermediate goods trade volume corresponding to the missing linkage in the industrial and supply chains as represented in the original MRIO table. From **a** to **d** are chord diagrams for the years 2019 to 2022, respectively.

countries, which enables Altasia to have a more significant substitution effect on China. Secondly, the risk exposure of CPN primarily relates to resource-intensive and labor-intensive industries such as “Agriculture, Hunting, Forestry and Fishing (S01)”, “Wood and Products of Wood and Cork (S06)”, “Real Estate Activities (S29)”. In contrast, technology-intensive industries like “Chemicals and Chemical Products (S09)” and “Electrical and Optical Equipment (S14)” are almost unaffected. This indicates that Southeast Asian countries are likely to undertake the production and processing segments of China’s lower and middle-end industries. Simultaneously, China is vigorously developing high-end manufacturing with higher technological content and value-added to replace the low-end excess capacity that is transferred to Southeast Asian countries.

Changes in the macroscopic topological structure of CPN. In this section, several network-level indicators of three types of GIVCBN models were calculated, respectively, and their ratios in the counterfactual model and null model were used to gauge the influence of Altasia and ASEAN on the CPN.

As shown in Fig. 13, the ratio of the APL in both counterfactual models to that in the null model is greater than 1, which indicates that the formation of Altasia will extend China’s industrial and supply chains, increase the distance between production and market endpoints, and decrease the efficiency of value creation through the circulation of intermediate goods. In comparison, if ASEAN can establish close cooperation with East and South Asian countries, it will not only directly lead to more of China’s industrial and supply chains relocating to Altasia economies. However, it will also indirectly deepen the separation among China’s internal industrial sectors.

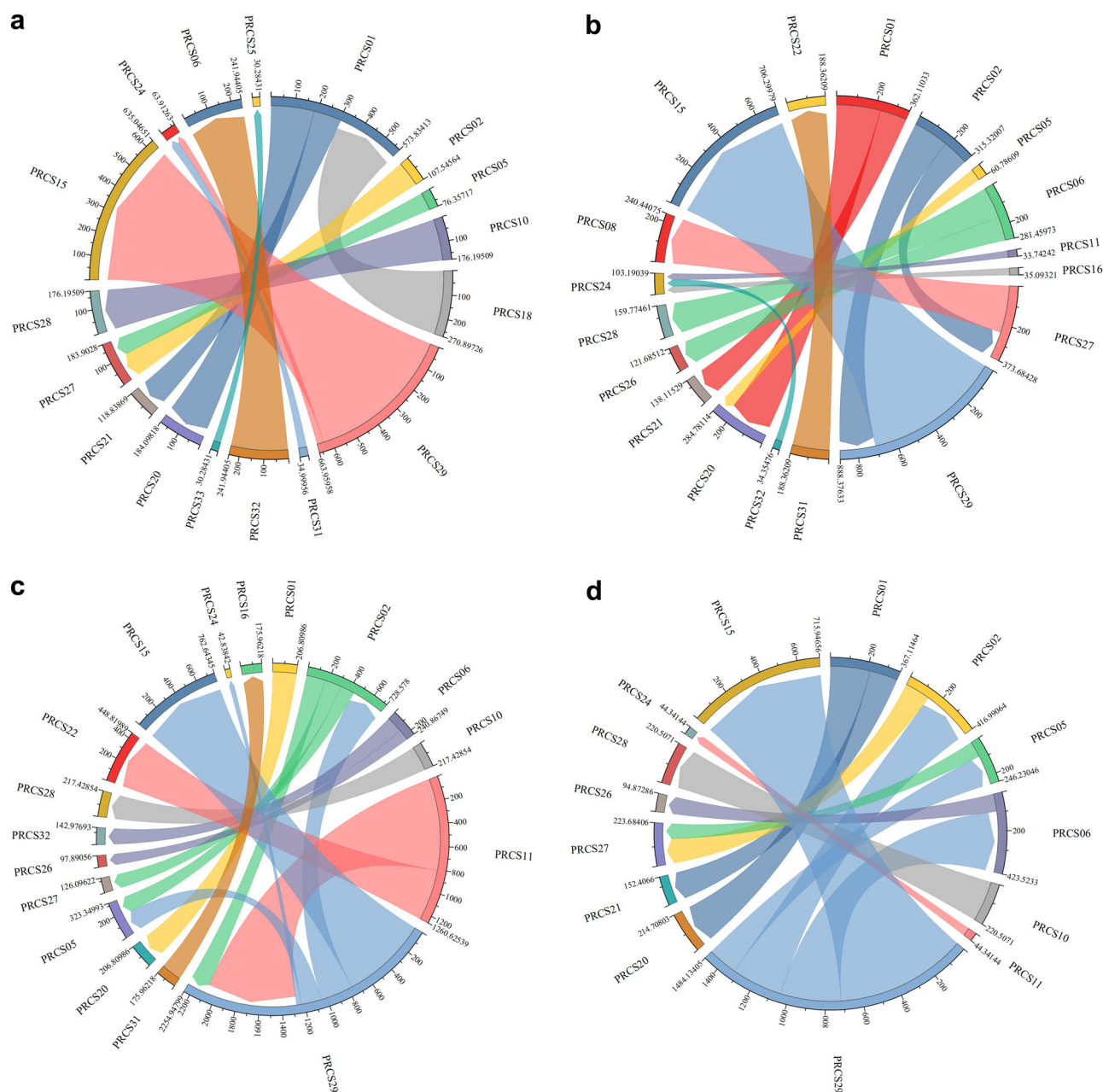


Fig. 12 Changes in the topological structure of CPN by comparing GIVCBN-I and GIVCBN-III models. From a to d are chord diagrams for the years 2019 to 2022, respectively.

Notably, the APL exhibits a fluctuating trend and has been continuously rebounding since 2018. This phenomenon indicates that China is adjusting its industrial layout in response to the GVC restructuring. On the one hand, China continuously strengthens economic and trade cooperation with ASEAN member countries through the RCEP, which is led by itself, exporting some of its excess capacity in resource-intensive and labor-intensive industries to Southeast Asian countries. On the other hand, China is also strategizing its overseas layouts through the international business strategy. According to a February 2024 analysis of trade data by the Financial Times, China is shipping more goods to the United States via Mexico to circumvent the high tariffs imposed by the Trump administration and retained by the Biden-led White House. Additionally, China has trade surpluses with Vietnam, Singapore, and the Philippines, and these countries' trade surpluses with the United States are also

expanding, indicating that Chinese manufacturers continue to benefit from the demand of American consumers for their products with competitive prices.

Figure 14 clearly illustrates that the ratios of (C) in the two scenarios are all lower than 1, indicating that Altasia is causing a significant break in the interconnection between some upstream and downstream sectors within the CPN. The GIVCBN-II model consistently shows lower results than the GIVCBN-III model, suggesting that the close cooperation among ASEAN countries is also contributing to the partial decoupling of China's industrial and supply chains. However, the impact of this is dwarfed by the formation of Altasia, which is causing a more substantial shift in the landscape.

With Southeast Asian countries as the majority, Altasia has become the biggest beneficiary of the fifth international industrial transfer due to the superposition of human costs and security

Table 4 Comparison between GIVCBN-II and GIVCBN-I models.

Upstream sector	Downstream sector							
	2007	2008	2009	2010	2011	2012	2013	2014
PRCS01	PRCS11 PRCS20 PRCS21	PRCS20 PRCS21 PRCS29	PRCS11 PRCS20 PRCS21 PRCS29	PRCS11 PRCS20 PRCS21 PRCS29	PRCS20 PRCS21	PRCS02	PRCS20 PRCS21	PRCS02 PRCS11 PRCS21 PRCS29
PRCS02	PRCS21 PRCS28	PRCS28	PRCS20 PRCS28	PRCS20 PRCS21 PRCS28	PRCS05 PRCS20 PRCS21 PRCS28	PRCS05 PRCS27 PRCS28 PRCS29	PRCS05 PRCS27 PRCS28 PRCS29	PRCS05 PRCS27 PRCS28 PRCS29
PRCS04	PRCS08	—	—	—	—	PRCS24	PRCS24	PRCS24
PRCS05	PRCS27	PRCS27	PRCS28	PRCS22 PRCS27	—	PRCS27	PRCS27	PRCS27
PRCS06	PRCS27	PRCS21 PRCS27	PRCS21	PRCS21	—	PRCS23 PRCS32	PRCS20 PRCS23 PRCS24 PRCS31 PRCS32	PRCS20 PRCS23 PRCS24 PRCS32
PRCS10	PRCS24 PRCS25	PRCS24 PRCS25	PRCS24	—	PRCS25	—	—	—
PRCS11	PRCS20 PRCS21 PRCS24 PRCS25 PRCS26 PRCS28	PRCS20 PRCS21 PRCS24 PRCS28 PRCS29	PRCS20 PRCS21 PRCS29	PRCS05 PRCS22 PRCS24 PRCS29	PRCS05 PRCS22 PRCS29	PRCS05 PRCS29	PRCS05 PRCS24	PRCS05 PRCS24
PRCS15	—	—	—	—	—	PRCS05	PRCS05	PRCS05
PRCS16	—	PRCS25	—	PRCS24	—	—	—	—
PRCS18	—	PRCS06	—	—	PRCS01	—	—	—
PRCS24	PRCS28 PRCS29	PRCS27 PRCS29	PRCS27 PRCS29	PRCS27 PRCS29	PRCS27 PRCS29	—	—	—
PRCS25	PRCS08	PRCS24	PRCS24	—	—	—	—	—
PRCS26	—	—	—	PRCS28	PRCS28	PRCS28	PRCS28	PRCS28
PRCS29	PRCS17	—	PRCS06	PRCS02	PRCS05 PRCS18	PRCS03	PRCS03	PRCS03
PRCS31	—	—	PRCS05 PRCS16	PRCS16 PRCS25	PRCS25	PRCS16	PRCS16	PRCS08
PRCS32	PRCS06 PRCS24	PRCS06	—	PRCS24	PRCS16 PRCS24	PRCS16	PRCS16	—
PRCS33	PRCS21	PRCS10	PRCS26	PRCS16	PRCS26	—	PRCS26	PRCS26
Sum	23	21	19	23	20	16	21	21
	2015	2016	2017	2018	2019	2020	2021	2022
PRCS01	PRCS02 PRCS21 PRCS29	PRCS02 PRCS21	PRCS02 PRCS11 PRCS21 PRCS29	PRCS11 PRCS21 PRCS29	PRCS11 PRCS20 PRCS21 PRCS29	PRCS11 PRCS20 PRCS21	PRCS20 PRCS29	PRCS20 PRCS21
PRCS02	PRCS28 PRCS29	—	—	PRCS01	PRCS27	PRCS27 PRCS29	PRCS05 PRCS27 PRCS29	PRCS27 PRCS29
PRCS04	PRCS24	PRCS24	PRCS24	PRCS24	—	—	—	PRCS08
PRCS05	PRCS27	PRCS20	PRCS27	—	PRCS27	PRCS20 PRCS27	—	PRCS08 PRCS27
PRCS06	PRCS17 PRCS20 PRCS23 PRCS24 PRCS28 PRCS31 PRCS32	PRCS20 PRCS23 PRCS28 PRCS32	PRCS20 PRCS23 PRCS32	PRCS17 PRCS20 PRCS24 PRCS26 PRCS28 PRCS32	PRCS17 PRCS26 PRCS32	PRCS17 PRCS26 PRCS28 PRCS31 PRCS32	PRCS17 PRCS26 PRCS32	PRCS17 PRCS26 PRCS32
PRCS07	—	PRCS25	PRCS25	—	—	—	—	—
PRCS10	—	PRCS24	PRCS24	PRCS28	PRCS24 PRCS28	PRCS24	PRCS28	PRCS24 PRCS28
PRCS11	PRCS05 PRCS24	PRCS05 PRCS23	PRCS05 PRCS23	PRCS23	PRCS05	PRCS05 PRCS24	PRCS05 PRCS22 PRCS24 PRCS29	PRCS05 PRCS22 PRCS24 PRCS29
PRCS15	PRCS05	PRCS05	PRCS05	PRCS05	PRCS05	PRCS05	PRCS05	PRCS05

Table 4 (continued)

Upstream sector	Downstream sector							
PRCS16	PRCS25	PRCS25	PRCS25	PRCS24	—	PRCS24	PRCS24	PRCS24
PRCS18	PRCS05	—	—	PRCS01	PRCS01	—	—	PRCS01
PRCS24	PRCS27	—	PRCS27	PRCS28	—	—	—	—
PRCS26	PRCS28	PRCS28	PRCS28	PRCS28	—	—	—	—
PRCS27	—	—	—	PRCS04	—	PRCS08	—	—
PRCS29	PRCS03	PRCS03	PRCS03	PRCS06	PRCS02	PRCS02	PRCS02	PRCS02
				PRCS02				
				PRCS05				
				PRCS06				
				PRCS15				
PRCS31	PRCS16	PRCS08	—	PRCS24	PRCS16	PRCS22	PRCS16	PRCS16
				PRCS08				
				PRCS24				
PRCS32	PRCS24	PRCS10	PRCS10	PRCS06	PRCS10	PRCS16	PRCS10	PRCS10
				PRCS08				
				PRCS17				
PRCS33	PRCS04	PRCS04	PRCS04	—	PRCS25	PRCS04	—	—
Sum	27	24	23	29	24	28	24	27

factors. According to relevant statistics, the total Foreign Direct Investment (FDI) attracted by Southeast Asian countries reached a record \$222.5 billion in 2022. The United States is the global leader in capital investment projects in Southeast Asia, spending \$74.3 billion to build factories and other projects between 2018 and 2022, mainly hoping to diversify its supply chain by reducing its over-reliance on China. Nevertheless, under the strategic influence of the “De-Sinicization” strategy implemented by the United States, China is not passively responding but actively relocating its industrial and supply chains to Southeast Asian countries or Mexico. During the same period, China’s investment also reached \$68.5 billion, aiming to maintain exports to the United States and Europe through Southeast Asian factories under the complex international background. Therefore, Altasia has become an essential broker between significant economies.

As a result, under the premise that China and the United States would jointly seek Altasia as a strategic partner, China’s position as the “World’s Factory” began to decline. This is specifically manifested in the industrial and supply chains of the CPN being less dense than in the past, which is indicated by a decrease in the network’s (C) value.

According to the definition, a higher value of ASY indicates a lower degree of vertical specialization in the division of labor within the production network. As shown in Fig. 15, the ratios of ASY in the two scenarios are all higher than 1. Moreover, in 2020, the ratio of the GIVCBN-II model to the GIVCBN-I model even exceeded 1.4. The fact behind this phenomenon is that the COVID-19 pandemic has severely impacted the coupling relationship between the CPN and the GPN, and significantly reduced China’s capacity to export intermediate goods to the global supply chains. Simultaneously, the substitution effect of Altasia for China has begun to manifest.

The asymmetrical nature of different countries’ responses to COVID-19 further complicated global supply chain issues. After the outbreak of the COVID-19 pandemic, China instituted one of the strictest containment strategies in the world, known as the “Zero COVID” approach. With production halted in one of the world’s most productive countries, COVID-19 shocked the global economy into widespread unpredictability and shortages. In this context, companies and governments are reconsidering their

reliance on “Made in China” and are beginning to diversify their supply sources to enhance resilience. At the same time, in response to the pandemic and its economic impacts, many countries have introduced policies and incentives aimed at attracting FDI and encouraging domestic manufacturing. Countries in Southeast Asia, Mexico, and India have implemented favorable tax regimes, subsidies, and infrastructure improvements, making them more competitive as alternative manufacturing locations.

Changes in the microscopic topological structure of CPN.

Although global economic integration has experienced a period of rapid development, the industrial and supply chains of Southeast Asia have long been positioned as “Semi-Periphery” within the GPN, maintaining a relatively independent internal structure. In recent years, the United States’ policy and strategy towards China have shifted, as evidenced by leveraging strategic frameworks such as the CPTPP and the IPEF to unite Southeast Asian countries. This move aims to turn them into new manufacturing hubs and proxies in the Asia-Pacific region, resulting in a trend of both active and passive relocation of industrial supply chains from China.

To quantitatively analyze the extent to which Altasia has weakened China’s *influence scope*, *profitability*, and *robustness of risk* within the global industrial and supply chains, this paper integrates the calculation results of five types of network centrality indicators from 2007 to 2022. In this section, the *Probability Density Curve (PDF)* was used to evaluate the overall impact on Altasia and China based on the *Mean (μ)* and *Standard Deviation (σ)* of the normal distribution.

Once Altasia becomes a tight economic organization, both its and China’s industrial sectors will see a reduced influence scope on the upstream ($\mu_3 = 0.835$ and $\mu_1 = 0.878$ in Fig. 16) and downstream ($\mu_3 = 0.772$ and $\mu_1 = 0.892$ in Fig. 17) sectors across all other economies. If ASEAN is considered solely as a tight core, then the negative impact of Altasia on China would be significantly reduced ($\mu_1 < \mu_2$ in Figs. 16 and 17), and the variability among the data points representing the influence scope of Altasia’s and China’s internal industrial sectors would decrease ($\sigma_4 < \sigma_3$ and $\sigma_2 < \sigma_1$ in Figs. 16 and 17). Obviously, the

Table 5 Comparison between GIVCBN-III and GIVCBN-I models.

Upstream sector	Downstream sector							
	2007	2008	2009	2010	2011	2012	2013	2014
PRCS01	—	PRCS29	PRCS11 PRCS20	PRCS11 PRCS20 PRCS21	PRCS20 PRCS21	—	PRCS20 PRCS21	PRCS11 PRCS21
PRCS02	PRCS28	—	PRCS28	PRCS20 PRCS21 PRCS28	PRCS20 PRCS21	PRCS05 PRCS28	PRCS05 PRCS28	PRCS05 PRCS27 PRCS28 PRCS29
PRCS04	—	—	—	—	—	PRCS24	—	—
PRCS05	—	—	PRCS28	PRCS22	—	PRCS27	PRCS27	PRCS27
PRCS06	PRCS27	—	PRCS21	PRCS21	—	—	PRCS24 PRCS32	—
PRCS10	PRCS24	PRCS24	—	—	—	—	—	—
PRCS11	PRCS28	PRCS24 PRCS28	PRCS20 PRCS21	PRCS24 PRCS29	PRCS22 PRCS29	PRCS29	PRCS24	—
PRCS16	—	PRCS25	—	—	—	—	—	—
PRCS18	—	PRCS06	—	—	—	—	—	—
PRCS24	—	—	—	PRCS27	—	—	—	—
PRCS26	—	—	—	—	—	—	PRCS28	PRCS28
PRCS29	—	—	—	—	PRCS18	PRCS03	PRCS03	PRCS03
PRCS31	—	—	—	PRCS16	—	PRCS16	PRCS31	PRCS08
PRCS32	PRCS06	PRCS06	—	—	PRCS24	—	—	—
PRCS33	—	—	PRCS26	—	PRCS26	—	PRCS26	PRCS26
Sum	5	7	8	12	9	7	12	11
	2015	2016	2017	2018	2019	2020	2021	2022
PRCS01	PRCS21	PRCS21	PRCS11 PRCS29	PRCS21	PRCS20 PRCS21	PRCS20 PRCS21	PRCS20	PRCS20 PRCS21
PRCS02	PRCS28 PRCS29	—	—	—	PRCS27	PRCS27 PRCS29	PRCS05 PRCS27 PRCS29	PRCS27
PRCS04	—	PRCS24	PRCS24	PRCS24	—	—	—	—
PRCS05	PRCS27	—	PRCS27	—	PRCS27	PRCS20	—	PRCS27
PRCS06	PRCS24 PRCS29 PRCS32	PRCS28 PRCS32	PRCS32	PRCS26 PRCS28	—	PRCS26 PRCS28	PRCS26 PRCS32	PRCS26
PRCS10	—	—	—	PRCS28	PRCS28	—	PRCS28	PRCS28
PRCS11	PRCS24	PRCS05	PRCS05	—	—	PRCS24	PRCS22 PRCS29	PRCS24
PRCS15	PRCS05	—	—	PRCS05	—	—	—	—
PRCS16	PRCS25	PRCS25	PRCS25	PRCS24	—	PRCS24	—	—
PRCS18	—	—	—	—	PRCS01	—	—	—
PRCS24	PRCS27	—	—	—	—	—	—	—
PRCS26	PRCS28	PRCS28	PRCS28	—	—	—	—	—
PRCS27	—	—	—	—	—	PRCS08	—	—
PRCS29	PRCS03	PRCS03	PRCS03	PRCS05 PRCS15	PRCS15 PRCS24	PRCS15	PRCS02 PRCS05 PRCS15 PRCS24 PRCS16	PRCS02 PRCS05 PRCS06 PRCS15
PRCS31	—	PRCS08	—	PRCS08	PRCS24	PRCS22	—	—
PRCS32	—	PRCS24	PRCS10 PRCS24	PRCS06	PRCS06	PRCS24	—	—
PRCS33	PRCS16 PRCS26	PRCS16 PRCS26	PRCS04 PRCS16 PRCS26	—	PRCS25	—	—	—
Sum	15	12	14	11	11	13	14	11

anti-globalization trend generated by ASEAN is weaker than that of Altasia.

Under the premise of Altasia as a tightly integrated economic organization, its and China's overall profitability of industrial sectors would see a slight decline ($\mu_3 = 0.966$ and $\mu_1 = 0.962$ in Fig. 18). Altasia's industrial sectors, dispersed across different economies, exhibit more significant variability among data points ($\sigma_3 = 0.493$ in Fig. 18). However, under the premise of a tight integration among ASEAN member countries, the overall

profitability of Altasia's industrial sectors shows a more noticeable decrease ($\mu_4 = 0.843$ in Fig. 18), while that of China slightly increases ($\mu_2 = 1.025$ in Fig. 18). This result demonstrates that by deepening economic and trade cooperation with ASEAN through the RCEP, China can gain incredible benefits in the global industrial and supply chains.

Overall, after the formation of either a complete (including ASEAN, East Asian, and South Asian) or partial (only ASEAN) tight industrial and supply chain coupling in Altasia, the overall

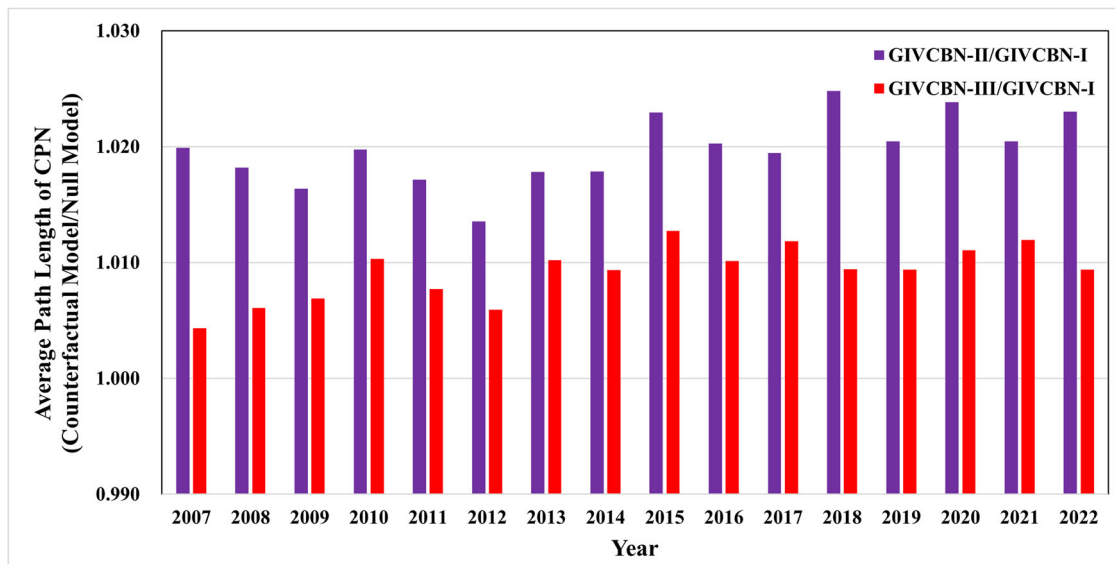


Fig. 13 Average path length of CPN. This figure shows the trend of APL changes in CPNs from 2007 to 2022, where the purple bars represent the ratio of APL values in CPN from the GIVCBN-II model to those in the GIVCBN-I model, and the red bars represent the ratio of APL values in CPN from the GIVCBN-III model to those in the GIVCBN-I model.

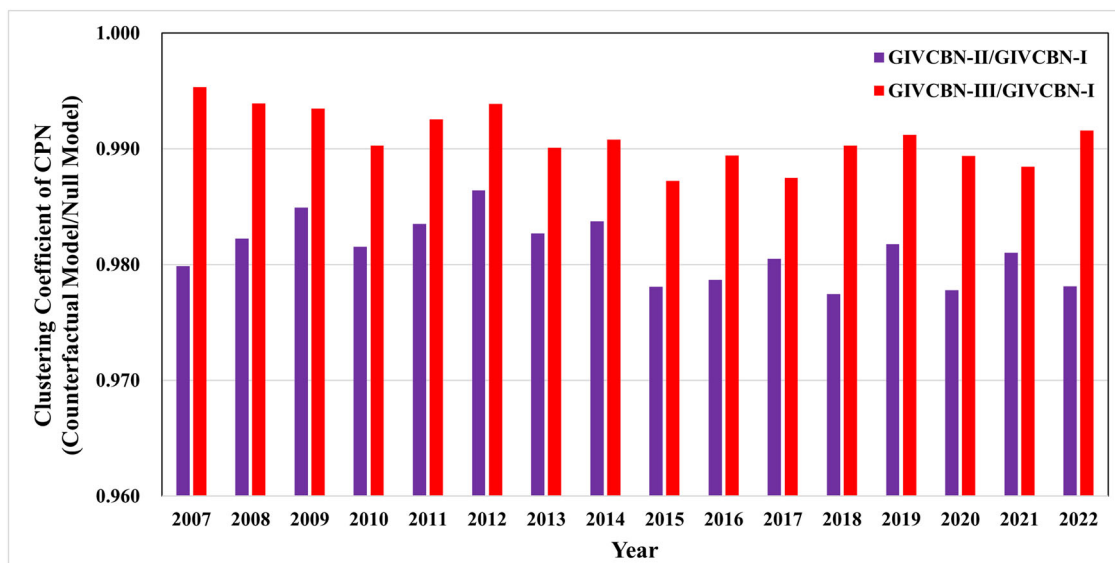


Fig. 14 Average clustering coefficient of CPN. This figure shows the trend of changes in CPNs from 2007 to 2022, where the purple bars represent the ratio of values in CPN from the GIVCBN-II model to those in the GIVCBN-I model, and the red bars represent the ratio of values in CPN from the GIVCBN-III model to those in the GIVCBN-I model.

robustness of China's industrial sectors against uncertainties from upstream ($\mu_1 = 0.973$ and $\mu_2 = 0.987$ in Fig. 19) and downstream ($\mu_1 = 0.978$ and $\mu_2 = 0.988$ in Fig. 20) do not show a significant decline. However, in the former scenario, there is more significant variability among the data points ($\sigma_1 > \sigma_2$ in Figs. 19 and 20), indicating that some industrial sectors experience different levels of impact. Even if Altasia prompts the relocation of industrial chains within China, China maintains close economic ties with other economies worldwide and leverages these connections to hedge against external risk shocks.

Changes in the specific industrial sectors of CPN. This paper selects seven industries as empirical analysis objects based on

their intensity of factors, including resource-intensive industries ["Agriculture, Hunting, Forestry and Fishing (PRCS01)", "Mining and Quarrying (PRCS02)", "Food, Beverages and Tobacco (PRCS03)"], labor-intensive industries ["Textiles and Textile Products (PRCS04)"], capital-intensive industries ["Basic Metals and Fabricated Metal (PRCS09)"], and technology-intensive industries ["Chemicals and Chemical Products (PRCS12)", "Electrical and Optical Equipment (PRCS14)"]. In this section, the boxplots are used to illustrate the fluctuation intervals of these factor-intensive industries.

Degree centrality measures a sector's direct industrial linkages in the production network. The in-degree centrality reflects the importance of an industrial sector as the demander of

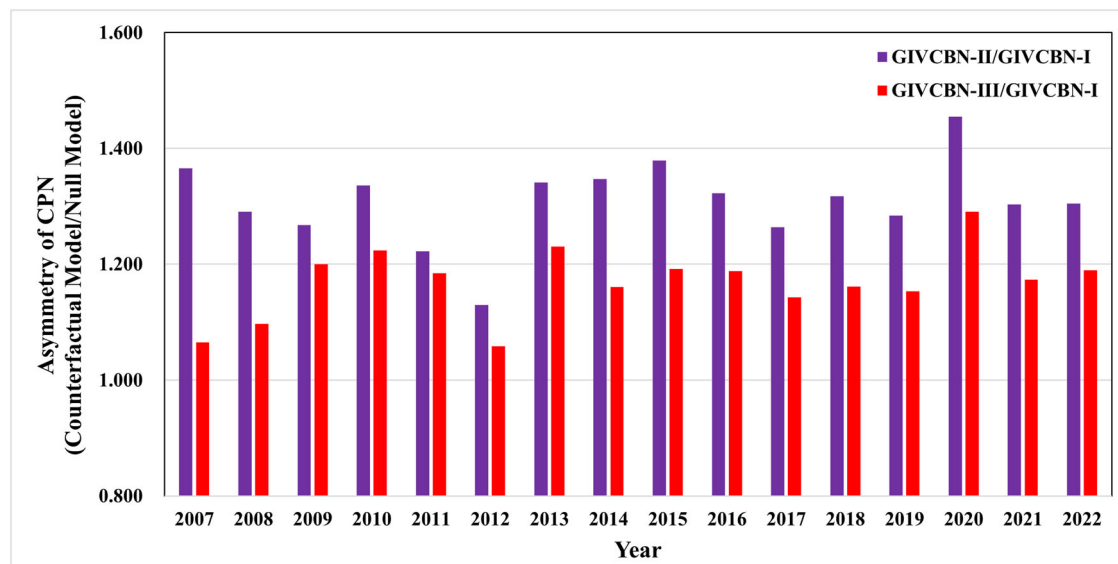


Fig. 15 Asymmetry of CPN. This figure shows the trend of ASY changes in CPNs from 2007 to 2022, where the purple bars represent the ratio of ASY values in CPN from the GIVCBN-II model to those in the GIVCBN-I model, and the red bars represent the ratio of ASY values in CPN from the GIVCBN-III model to those in the GIVCBN-I model.

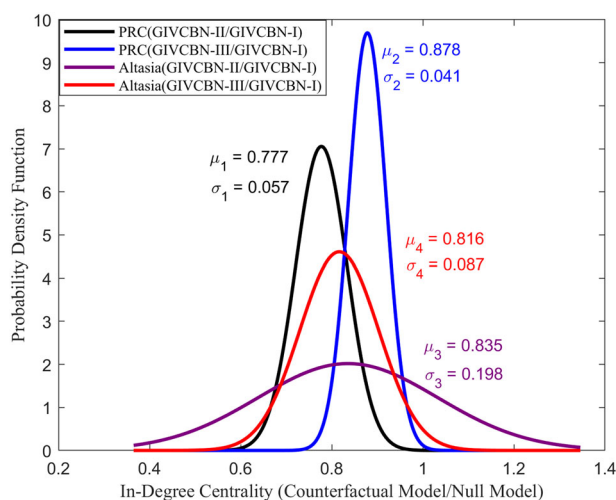


Fig. 16 Probability density function of in-degree centrality. The probability density function curve is drawn based on data from all relevant industrial sectors from 2007 to 2022, reflecting the potential impact capacity of Altasia and ASEAN on China over more than a decade.

intermediate goods. In contrast, the out-degree centrality reflects the importance of a sector as the supplier of intermediate goods. Altasia's industrial substitution effect on China shows a certain regularity from the in-degree or out-degree centrality perspective, as shown in Figs. 21 and 22. First, the ratios of the three resource-intensive industries are significantly less than 1, indicating that Altasia economies have seriously weakened China's influence scope in these fields. Second, Altasia's direct impacts on China's capital-intensive and technology-intensive industries are relatively small, especially when the latter's "Electrical and Optical Equipment (PRCS14)" sector supplies intermediate goods. Altasia will only partially replace the functions and positions of these factor-intensive industries in China. Third, the ratio of "Agriculture, Hunting, Forestry and Fishing (PRCS01)" is the smallest when China plays the role of supplier, indicating that Altasia has

replaced China as the main export region for such intermediate goods.

From an overall perspective, compared with Altasia, ASEAN has a weaker industrial substitution effect on China, which indicates that economies such as Taiwan, India, Japan, and South Korea play an essential role in Altasia. Taking "Electrical and Optical Equipment" as an example, Taiwan, South Korea, and Japan own these core upstream sectors that provide critical intermediate goods for developing China's electronic equipment manufacturing industry. Taking India as another example, based on the window period of global geopolitical tensions, India has attracted foreign capital to set up factories. Since 2020, India's electronic products' average annual export growth rate has been 30.4%, and its importance as a global electronic products trade node is rising. In addition, in the context of global value chain restructuring and COVID-19, Japan's critical industrial and supply chains are broken, which promotes strengthening the critical industrial chain's localization and starting the supply chain reform—the "China + 1" strategy. However, it is challenging to replicate a mature manufacturing industry chain and supply chain quickly, so this strategy is not easy to achieve.

Betweenness centrality measures the extent to which an industrial sector acts as a bridge for the flow of products and services in the GPN, and its ratio of the counterfactual model to the null model for different industrial sectors reflects the extent to which their hub function is disturbed. From the perspective of betweenness centrality, the ratios of six representative sectors in China are all less than 1, indicating that they are affected by the industrial substitution effect of a hypothetical economic entity, as shown in Fig. 23. Generally speaking, the most vital industrial substitution effect is mainly in the resource-intensive industry, among which the industrial relocation trend of "Agriculture, Hunting, Forestry and Fishing (PRCS01)" is the most obvious. In contrast, the ratio of "Electrical and Optical Equipment (PRCS14)" exceeds 1, reflecting that Altasia not only does not weaken PRCS14's hub function but actually plays a significant role in enhancing its profitability in the GPN. This underscores the dynamic nature of industrial substitution effect and its potential to reshape the global production landscape.

Taking ASEAN as a whole, the ratios of most industrial sectors are greater than 1, especially the capital-intensive and

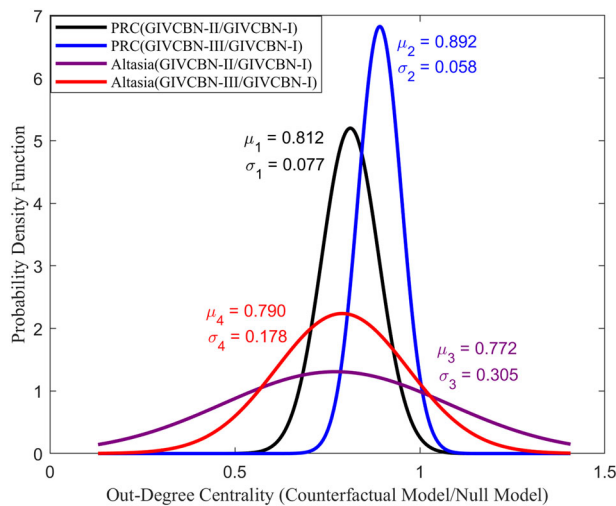


Fig. 17 Probability density function of out-degree centrality. The settings in this figure are the same as those in Fig. 16.

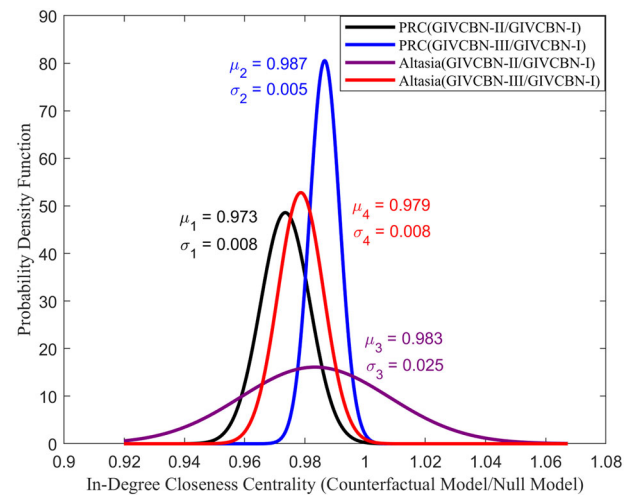


Fig. 19 Probability density function of in-degree closeness centrality. The settings in this figure are the same as those in Fig. 16.

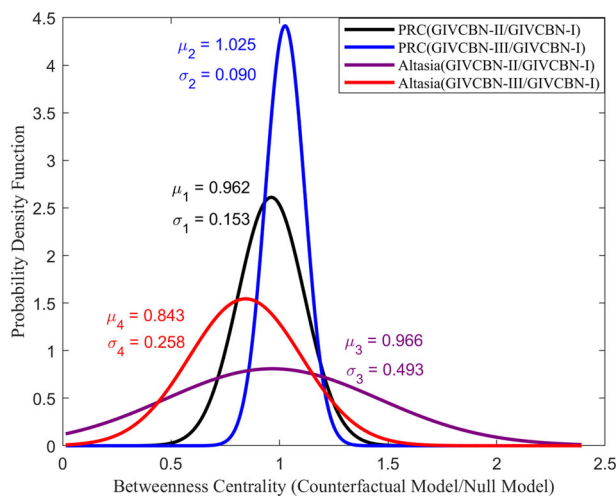


Fig. 18 Probability density function of betweenness centrality. The settings in this figure are the same as those in Fig. 16.

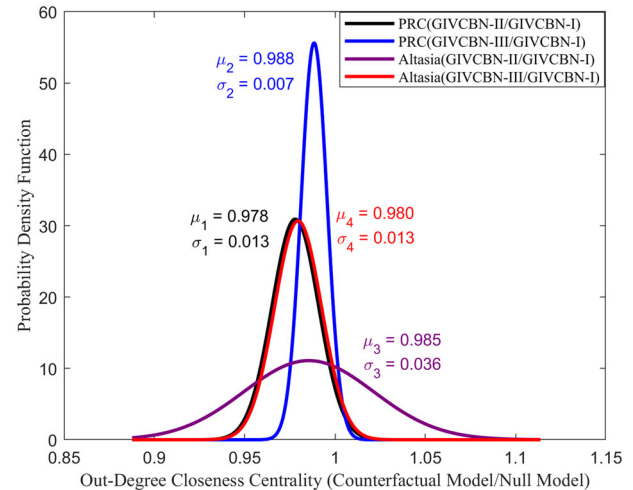


Fig. 20 Probability density function of out-degree closeness centrality. The settings in this figure are the same as those in Fig. 16.

technology-intensive industries, indicating that ASEAN will strengthen the profitability of these sectors in China. In recent years, China has taken the initiative to transfer some low-value-added and labor-intensive industrial sectors, such as clothing, toys, mechanical and electrical products processing and assembly, to Vietnam, India and other Asian neighboring economies, while increasing investment in higher value-added and more technological innovation linkages in the industrial chain, so as to provide intermediate goods with higher value for Southeast Asia. Since Japan and South Korea's scientific and technological strengths are in the leading position in the world (such as South Korea's semiconductor and Japan's precision machine tools), Altasia has a relatively significant substitution effect for China's technology-intensive industry, while ASEAN mainly undertakes low-end sectors, and even rely on China's industrial and supply chains to a large extent.

Closeness centrality, a measure used in network analysis, quantifies the extent to which an industrial sector is central to the GPN. In our analysis, the ratio of the counterfactual model to the null model is close to 1, indicating that China's status as a global production center has not been affected by the apparent spillover

effect of other economies in the region, as shown in Figs. 24 and 25. First of all, China has the world's largest manufacturing industry and the most complete industrial chain, which can independently complete the whole process from raw material procurement to manufacturing, and then to sales and service, making it impossible for enterprises from other countries or regions to truly implement the so-called "China exit strategy". Second, China's dominance in the high-tech field is not a recent development. Despite attempts by the United States to limit investment in China's science and technology, American high-tech enterprises like Apple and Boeing still heavily depend on China's industrial and supply chain. This is a testament to China's long-standing technological prowess. It has taken over three decades for China to become a global production center, and the notion of any region replacing it overnight is unrealistic. Third, China has more advantages than other economies in terms of the business environment, such as its robust legal system and favorable tax policies, and top talents, with a large pool of highly skilled workers and a strong emphasis on education. In addition, the high degree of integration and deep cooperation with the industrial chain of some economies (such as Japan and South Korea) makes it more challenging for these economies to shake China's central position.

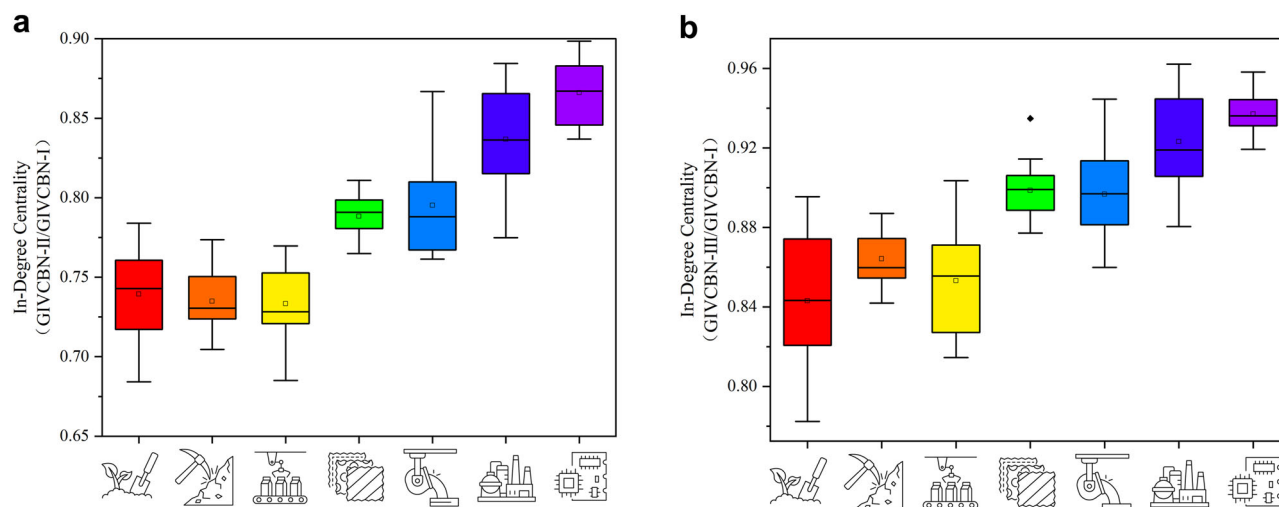


Fig. 21 Statistics on the in-degree centrality of industrial sectors in the GPN. The boxplots provide a visualization of the median line, reflecting the extent of the industrial substitution effect caused by Altasia (GIVCBN-II/GIVCBN-I) or ASEAN (GIVCBN-III/GIVCBN-I) on China. The smaller the ratio, the stronger the substitution effect. Besides, the icons on the horizontal axis in the boxplots represent PRCS01, PRCS02, PRCS03, PRCS04, PRCS09, PRCS12, PRCS14 sectors, respectively. **a** Temporal distribution of in-degree centrality (GIVCBN-II/GIVCBN-I). **b** Temporal distribution of in-degree centrality (GIVCBN-III/GIVCBN-I).

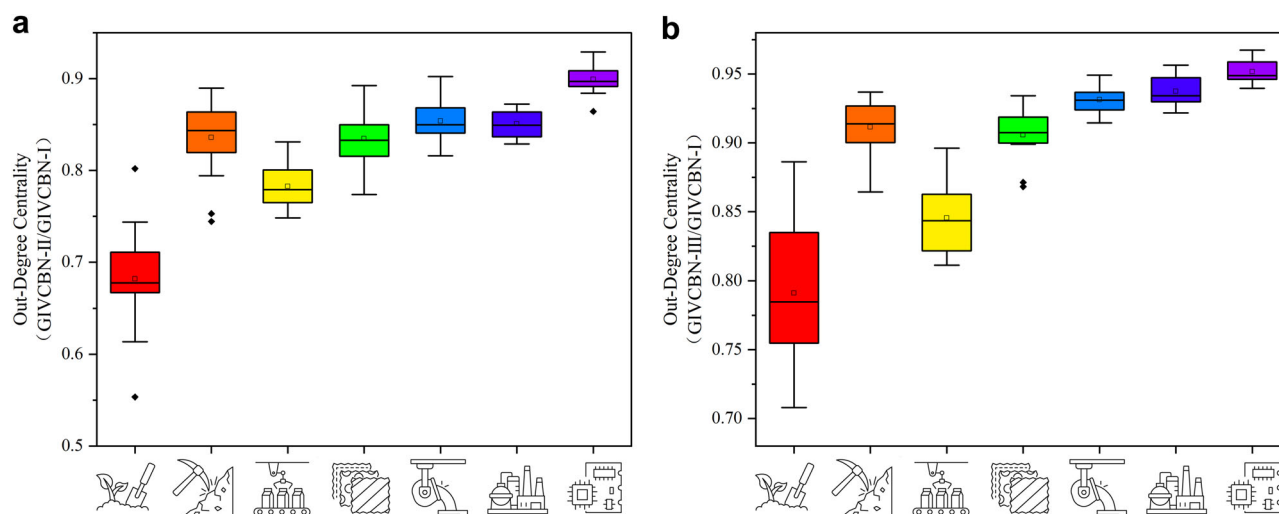


Fig. 22 Statistics on the out-degree centrality of industrial sectors in the GPN. **a** Temporal distribution of out-degree centrality (GIVCBN-II/GIVCBN-I). **b** Temporal distribution of out-degree centrality (GIVCBN-III/GIVCBN-I).

Discussion and conclusion

If, under the influence of geopolitics, ASEAN can form closer economic and trade cooperation with East Asia and South Asia, then the resulting Altasia will indeed affect the security of China's industrial and supply chains, particularly having a noticeable substitution effect on resource-intensive and labor-intensive industries. Therefore, we can see that China is accelerating the construction of the "Dual Circulation" new development pattern, aiming to enhance the resilience and security of industrial and supply chains, and achieve long-term sustainable development. To address the industrial substitution effect caused by Altasia on China, this study proposes specific policy recommendations as follows:

- (1) China should start to deepen trade and investment ties with countries that the United States prioritizes for nearshoring and offshoring, such as Vietnam, Thailand, and Mexico. These countries often require Chinese equipment and components for processing and production, while China
- (2) For resource-intensive industries at the highest risk of industrial relocation, relevant trade policies should be introduced to prevent excessive supply chain outsourcing and avoid the phenomenon of "Hollowing Out" in China.

also needs their resources and markets. Supporting Chinese manufacturing enterprises to increase overseas investment can effectively respond to the profound adjustments in the global industrial and supply chains by the United States and significant changes in international trade, thus mitigating the adverse impacts of "De-Risking" strategies on Chinese enterprises. At the same time, China should continue to increase investments in domestic high-tech and high-value-added manufacturing industries, providing Southeast Asia and Mexico with intermediate products of higher added value. In the trend towards diversified supply chains, China should achieve win-win cooperation with these countries to jointly promote the optimization and development of the global industrial and supply chains.

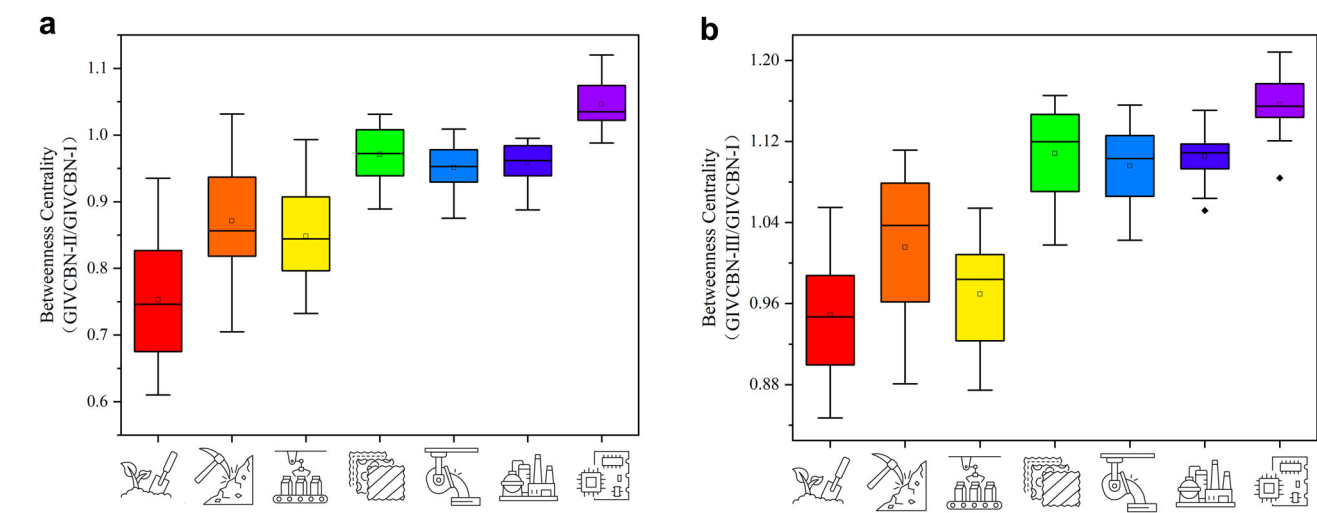


Fig. 23 Statistics on the betweenness centrality of industrial sectors in the GPN. a Temporal distribution of betweenness centrality (GIVCBN-II/GIVCBN-I). **b** Temporal distribution of betweenness centrality (GIVCBN-III/GIVCBN-I).

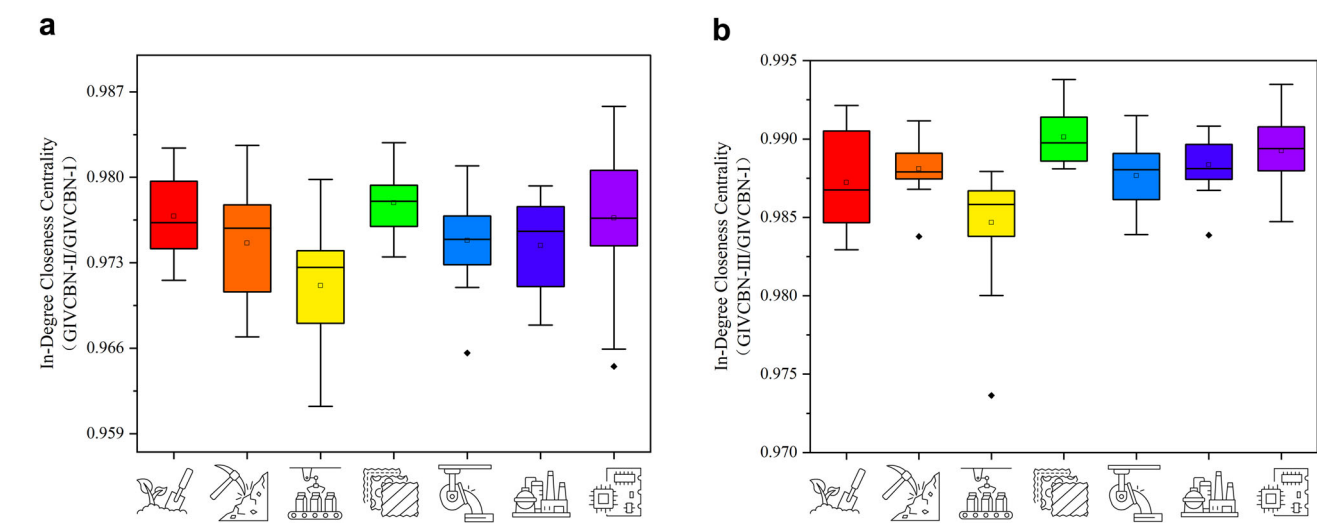


Fig. 24 Statistics on the in-closeness centrality of industrial sectors in the GPN. a Temporal distribution of in-degree closeness centrality (GIVCBN-II/GIVCBN-I). **b** Temporal distribution of in-degree closeness centrality (GIVCBN-III/GIVCBN-I).

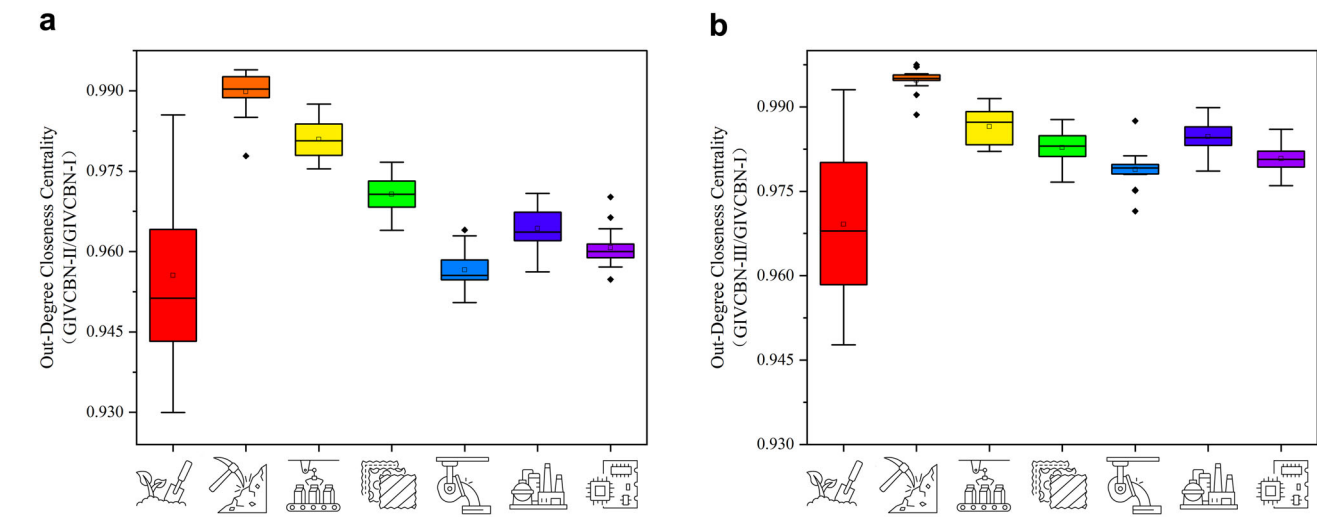


Fig. 25 Statistics on the out-closeness centrality of industrial sectors in the GPN. a Temporal distribution of out-degree closeness centrality (GIVCBN-II/GIVCBN-I). **b** Temporal distribution of out-degree closeness centrality (GIVCBN-III/GIVCBN-I).

For technology-intensive industries, China should strengthen the substitution of domestic products for imported ones, promoting the manufacturing industry's high-end, intelligent, and green development. It should actively cultivate strategic emerging industries such as new energy, new materials, advanced manufacturing, and electronic information, and proactively plan for future industries like brain-like intelligence, quantum information, deep-sea space, hydrogen energy, and energy storage to promote modern industrial clusters and accelerate the formation of new productive forces. At the same time, to enhance the local resilience of China's industrial and supply chains, further improvements should be made in the foreign investment environment, strengthening intellectual property protection and reducing market access barriers to attract multinational corporations to stay in China.

- (3) Actively cultivate regional and domestic value chains to build an independent, controllable, safe, and reliable complete industry chain system that can effectively withstand shocks from domestic and international contingencies, emergencies, and adverse effects, avoiding disruptions, blockages, and even breaks in the supply chain whenever possible. China should seize the post-pandemic window of opportunity for economic recovery and rely on the Belt and Road Initiative to build "China+X" industrial chains, implementing a more proactive and open strategy. By utilizing RCEP to construct a regionally dominated industrial chain network and actively striving to join CPTPP, China can more proactively expand international exchanges and cooperation, gaining more influence, decision-making power, and initiative in restructuring the global value chain. Through active participation in the reform and construction of the global governance system, China can contribute more Chinese wisdom, solutions, and strength to restructuring the global value chain, continuously enhancing China's influence and rule-making capacity in global governance.

Research outlook. Due to the dominance of processing trade by foreign enterprises, which is the main mode of foreign trade for emerging economies (such as China), it plays a crucial role in the development of the national economy. In recent years, the large-scale capacity transfer by foreign enterprises has threatened the industrial security of many countries and regions. MRIO tables contain rich inter-regional industrial and supply chain linkage information, providing a basis to judge the direction and intensity of industrial shifts on the GVC. However, most studies based on MRIO tables follow the "territorial" rules of national economic accounting, which treat the value-added activities of multinational corporations' branches within the host country the same as those of domestic enterprises, leading to inaccurate assessments of the negative impacts of multinational corporations' capacity transfers on the industrial structures of specific economies. Therefore, it is not only a necessity but also a promising opportunity to establish an IO model that reflects the relationships between domestic and foreign enterprises, and scholars have already made beneficial attempts in this direction, paving the way for more accurate and comprehensive assessments of the global industrial landscape.

In 2019, the OECD released the OECD-AMNE database, which, using the bilateral distribution matrix of multinational corporations' outputs, transforms the investment returns in the host countries' territorial added value to the parent country's entitled added value. This has become the optimal approach for depicting GVC activities under the conditions of multinational

corporations' influence, fully reflecting the adverse effects of the anti-globalization process on multinational corporations and their production networks in their respective countries. The data structure of the OECD-AMNE database's bilateral distribution matrix is a typical Multiple-Input Multiple-Output (MIMO) network, thus these types of MRIO tables can be referred to as Multi-Regional Multiple-Input Multiple-Output (MRMIMO) tables. Research utilizing the OECD-AMNE database has begun to show preliminary results (Cadestin et al. 2018; Andrenelli et al. 2018; Zhu et al. 2022; Duan and Cai 2022).

In light of the current clear trend of anti-globalization and the unstable geopolitical landscape, to further understand the extent to which multinational corporations' diversified layout strategies are reshaping the global industrial division system, and their impact on the security and resilience of GPN, it is necessary to distinguish between domestic and foreign enterprises within the industrial sectors. Therefore, in future research, we will establish a type of heterogeneous network based on MRMIMO tables that differentiates corporate ownership and the upstream and downstream relationships in the industrial and supply chains. We will measure the impact of foreign enterprises' capacity transfers on the resilience of GPN using various research methods such as time series analysis, counterfactual analysis, and game theory analysis to assess future trends.

Data availability

The datasets analyzed during the current study are available in the ADB Key Indicators Database (KIDB) repository, <https://kidb.adb.org/globalization>. And the datasets generated during the current study are available from the corresponding author on reasonable request.

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Author contributions

For greater transparency, the individual contributions of the authors are as follows: LZX: contributed to conceptualization, methodology, project administration, supervision; SJ: contributed to methodology, validation, formal analysis, investigation, resources, data curation, writing—review & editing, visualization; SMY: contributed to methodology, validation, data curation, writing—original draft, visualization; FKL: contributed to writing—review & editing, visualization.

Competing interests

The authors declare no competing interests.

Ethical approval

Ethical approval was not required as the study did not involve human participants.

Informed consent

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Additional information

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